

Q1. The shortest distance between the lines $\frac{x-3}{4} = \frac{y+7}{-11} = \frac{z-1}{5}$ and $\frac{x-5}{3} = \frac{y-9}{-6} = \frac{z+2}{1}$ is: **09 Apr 2024 (M)**

- (1) $\frac{178}{\sqrt{563}}$ (2) $\frac{187}{\sqrt{563}}$
 (3) $\frac{185}{\sqrt{563}}$ (4) $\frac{179}{\sqrt{563}}$

Q2. The square of the distance of the image of the point $(6, 1, 5)$ in the line $\frac{x-1}{3} = \frac{y}{2} = \frac{z-2}{4}$, from the origin is _____ **09 Apr 2024 (E)**

Q3. Let the line L intersect the lines $x - 2 = -y = z - 1$, $2(x + 1) = 2(y - 1) = z + 1$ and be parallel to the line $\frac{x-2}{3} = \frac{y-1}{1} = \frac{z-2}{2}$. Then which of the following points lies on L ? **09 Apr 2024 (M)**

- (1) $(-\frac{1}{3}, 1, -1)$ (2) $(-\frac{1}{3}, -1, 1)$
 (3) $(-\frac{1}{3}, 1, 1)$ (4) $(-\frac{1}{3}, -1, -1)$

Q4. Consider the line L passing through the points $(1, 2, 3)$ and $(2, 3, 5)$. The distance of the point $(\frac{11}{3}, \frac{11}{3}, \frac{19}{3})$ from the line L along the line $\frac{3x-11}{2} = \frac{3y-11}{1} = \frac{3z-19}{2}$ is equal to **09 Apr 2024 (E)**

- (1) 6 (2) 5
 (3) 4 (4) 3

Q5. If the shortest distance between the lines

$$L_1 : \vec{r} = (2 + \lambda)\hat{i} + (1 - 3\lambda)\hat{j} + (3 + 4\lambda)\hat{k}, \quad \lambda \in \mathbb{R}$$

$$L_2 : \vec{r} = 2(1 + \mu)\hat{i} + 3(1 + \mu)\hat{j} + (5 + \mu)\hat{k}, \quad \mu \in \mathbb{R}$$

is $\frac{m}{\sqrt{n}}$, where $\gcd(m, n) = 1$, then the value of $m + n$ equals **08 Apr 2024 (M)**

- (1) 390 (2) 384
 (3) 377 (4) 387

Q6. If the shortest distance between the lines $\frac{x-\lambda}{2} = \frac{y-4}{3} = \frac{z-3}{4}$ and $\frac{x-2}{4} = \frac{y-4}{6} = \frac{z-7}{8}$ is $\frac{13}{\sqrt{29}}$, then a value of λ is _____ **08 Apr 2024 (E)**

- (1) -1 (2) $-\frac{13}{25}$
 (3) $\frac{13}{25}$ (4) 1

Q7. Let $P(\alpha, \beta, \gamma)$ be the image of the point $Q(1, 6, 4)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Then $2\alpha + \beta + \gamma$ is equal to _____ **08 Apr 2024 (E)**

Q8. Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy -plane is the point Q . Let $OP = \gamma$; the angle between OQ and the positive x -axis be θ ; and the angle between OP and the positive z -axis be ϕ , where O is the origin. Then the distance of P from the x -axis is **08 Apr 2024 (M)**

- (1) $\gamma\sqrt{1 - \sin^2 \phi \cos^2 \theta}$ (2) $\gamma\sqrt{1 - \sin^2 \theta \cos^2 \phi}$
 (3) $\gamma\sqrt{1 + \cos^2 \phi \sin^2 \theta}$ (4) $\gamma\sqrt{1 + \cos^2 \theta \sin^2 \phi}$

Q9. The shortest distance between the lines $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ and $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ is **06 Apr 2024 (M)**

- (1) $8\sqrt{3}$ (2) $4\sqrt{3}$
 (3) $5\sqrt{3}$ (4) $6\sqrt{3}$

Q10. Let P be the point $(10, -2, -1)$ and Q be the foot of the perpendicular drawn from the point $R(1, 7, 6)$ on the line passing through the points $(2, -5, 11)$ and $(-6, 7, -5)$. Then the length of the line segment PQ is equal to

06 Apr 2024 (M)

Q11. Let $P(\alpha, \beta, \gamma)$ be the image of the point $Q(3, -3, 1)$ in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point $(2, 5, -1)$. If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to :

06 Apr 2024 (E)

(1) 36

(2) 81

(3) 72

(4) 18

Q12. If the shortest distance between the lines $\frac{x-\lambda}{3} = \frac{y-2}{-1} = \frac{z-1}{1}$ and $\frac{x+2}{-3} = \frac{y+5}{2} = \frac{z-4}{4}$ is $\frac{44}{\sqrt{30}}$, then the largest possible value of $|\lambda|$ is equal to _____

06 Apr 2024 (E)

Q13. If the line $\frac{2-x}{3} = \frac{3y-2}{4\lambda+1} = 4-z$ makes a right angle with the line $\frac{x+3}{3\mu} = \frac{1-2y}{6} = \frac{5-z}{7}$, then $4\lambda + 9\mu$ is equal

to :

05 Apr 2024 (M)

(1) 4

(2) 13

(3) 5

(4) 6

Q14. Let d be the distance of the point of intersection of the lines $\frac{x+6}{3} = \frac{y}{2} = \frac{z+1}{1}$ and $\frac{x-7}{4} = \frac{y-9}{3} = \frac{z-4}{2}$ from the point $(7, 8, 9)$. Then $d^2 + 6$ is equal to :

05 Apr 2024 (M)

(1) 69

(2) 78

(3) 72

(4) 75

Q15. Let (α, β, γ) be the image of the point $(8, 5, 7)$ in the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{5}$. Then $\alpha + \beta + \gamma$ is equal to :

05 Apr 2024 (E)

(1) 16

(2) 20

(3) 14

(4) 18

Q16. Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ and $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to _____

05 Apr 2024 (E)

Q17. Consider a line L passing through the points $P(1, 2, 1)$ and $Q(2, 1, -1)$. If the mirror image of the point $A(2, 2, 2)$ in the line L is (α, β, γ) , then $\alpha + \beta + 6\gamma$ is equal to _____

04 Apr 2024 (E)

Q18. Let the point, on the line passing through the points $P(1, -2, 3)$ and $Q(5, -4, 7)$, farther from the origin and at distance of 9 units from the point P , be (α, β, γ) . Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

04 Apr 2024 (M)

(1) 165

(2) 160

(3) 155

(4) 150

Q19. Let P be the point of intersection of the lines $\frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1}$ and $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-3}{2}$. Then, the shortest distance of P from the line $4x = 2y = z$ is _____

04 Apr 2024 (E)

(1) $\frac{5\sqrt{14}}{7}$ (2) $\frac{3\sqrt{14}}{7}$ (3) $\frac{\sqrt{14}}{7}$ (4) $\frac{6\sqrt{14}}{7}$

Q20. If the shortest distance between the lines $\frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$ and $\frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}$ k, and

$\int_0^k [x^2] dx = \alpha - \sqrt{\alpha}$, where $[x]$ denotes the greatest integer function, then $6\alpha^3$ is equal to _____

04 Apr 2024 (M)

Q21. If the shortest distance between the lines $\frac{x-\lambda}{-2} = \frac{y-2}{1} = \frac{z-1}{1}$ and $\frac{x-\sqrt{3}}{1} = \frac{y-1}{-2} = \frac{z-2}{1}$ is 1, then the sum of all possible values of λ is **01 Feb 2024 (M)**

- (1) 0 (2) $2\sqrt{3}$
(3) $3\sqrt{3}$ (4) $-2\sqrt{3}$

Q22. If the mirror image of the point $P(3, 4, 9)$ in the line $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-2}{1}$ is (α, β, γ) , then $14(\alpha + \beta + \gamma)$ is: **01 Feb 2024 (E)**

- (1) 102 (2) 138
(3) 108 (4) 132

Q23. Let the line of the shortest distance between the lines $L_1 : \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $L_2 : \vec{r} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(\hat{i} + \hat{j} - \hat{k})$ intersect L_1 and L_2 at P and Q respectively. If (α, β, γ) is the midpoint of the line segment PQ , then $2(\alpha + \beta + \gamma)$ is equal to **01 Feb 2024 (M)**

Q24. Let P and Q be the points on the line $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$ which are at a distance of 6 units from the point $R(1, 2, 3)$. If the centroid of the triangle PQR is (α, β, γ) , then $\alpha^2 + \beta^2 + \gamma^2$ is: **01 Feb 2024 (E)**

- (1) 26 (2) 36
(3) 18 (4) 24

Q25. Let (α, β, γ) be mirror image of the point $(2, 3, 5)$ in the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Then $2\alpha + 3\beta + 4\gamma$ is equal to **31 Jan 2024 (E)**

- (1) 32 (2) 33
(3) 31 (4) 34

Q26. The distance of the point $Q(0, 2, -2)$ from the line passing through the point $P(5, -4, 3)$ and perpendicular to the lines $\vec{r} = (-3\hat{i} + 2\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 5\hat{k})$, $\lambda \in \mathbb{R}$ and $\vec{r} = (\hat{i} - 2\hat{j} + \hat{k}) + \mu(-\hat{i} + 3\hat{j} + 2\hat{k})$, $\mu \in \mathbb{R}$ is **31 Jan 2024 (M)**

- (1) $\sqrt{86}$ (2) $\sqrt{20}$
(3) $\sqrt{54}$ (4) $\sqrt{74}$

Q27. The shortest distance between lines L_1 and L_2 , where $L_1 : \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+4}{2}$ and L_2 is the line passing through the points $A(-4, 4, 3)$, $B(-1, 6, 3)$ and perpendicular to the line $\frac{x-3}{-2} = \frac{y}{3} = \frac{z-1}{1}$, is **31 Jan 2024 (E)**

- (1) $\frac{121}{\sqrt{221}}$ (2) $\frac{24}{\sqrt{117}}$
(3) $\frac{141}{\sqrt{221}}$ (4) $\frac{42}{\sqrt{117}}$

Q28. A line passes through $A(4, -6, -2)$ and $B(16, -2, 4)$. The point $P(a, b, c)$ where a, b, c are non-negative integers, on the line AB lies at a distance of 21 units, from the point A . The distance between the points $P(a, b, c)$ and $Q(4, -12, 3)$ is equal to **31 Jan 2024 (E)**

Q29. Let Q and R be the feet of perpendiculars from the point $P(a, a, a)$ on the lines $x = y, z = 1$ and $x = -y, z = -1$ respectively. If $\angle QPR$ is a right angle, then $12a^2$ is equal to **31 Jan 2024 (M)**

Q30. Let (α, β, γ) be the foot of perpendicular from the point $(1, 2, 3)$ on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. then $19(\alpha + \beta + \gamma)$ is equal to : 30 Jan 2024 (M)

- (1) 102 (2) 101
(3) 99 (4) 100

Q31. If d_1 is the shortest distance between the lines $x + 1 = 2y = -12z$, $x = y + 2 = 6z - 6$ and d_2 is the shortest distance between the lines $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$, $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$, then the value of $\frac{32\sqrt{3}d_1}{d_2}$ is : 30 Jan 2024 (M)

Q32. Let $L_1 : \vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k})$, $\lambda \in R$, $L_2 : \vec{r} = (\hat{j} - \hat{k}) + \mu(3\hat{i} + \hat{j} + p\hat{k})$, $\mu \in R$ and $L_3 : \vec{r} = \delta(\hat{l} + m\hat{j} + n\hat{k})$, $\delta \in R$ be three lines such that L_1 is perpendicular to L_2 and L_3 is perpendicular to both L_1 and L_2 . Then the point which lies on L_3 is 30 Jan 2024 (E)

- (1) $(-1, 7, 4)$ (2) $(-1, -7, 4)$
(3) $(1, 7, -4)$ (4) $(1, -7, 4)$

Q33. Let a line passing through the point $(-1, 2, 3)$ intersect the lines $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ at $M(\alpha, \beta, \gamma)$ and $L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4}$ at $N(a, b, c)$. Then the value of $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ equals 30 Jan 2024 (E)

Q34. Let $P(3, 2, 3)$, $Q(4, 6, 2)$ and $R(7, 3, 2)$ be the vertices of ΔPQR . Then, the angle $\angle QPR$ is 29 Jan 2024 (E)

- (1) $\frac{\pi}{6}$ (2) $\cos^{-1}\left(\frac{7}{18}\right)$
(3) $\cos^{-1}\left(\frac{1}{18}\right)$ (4) $\frac{\pi}{3}$

Q35. Let O be the origin and the position vector of A and B be $2\hat{i} + 2\hat{j} + \hat{k}$ and $2\hat{i} + 4\hat{j} + 4\hat{k}$ respectively. If the internal bisector of $\angle AOB$ meets the line AB at C , then the length of OC is 29 Jan 2024 (M)

- (1) $\frac{2}{3}\sqrt{31}$ (2) $\frac{2}{3}\sqrt{34}$
(3) $\frac{3}{4}\sqrt{34}$ (4) $\frac{3}{2}\sqrt{31}$

Q36. Let PQR be a triangle with $R(-1, 4, 2)$. Suppose $M(2, 1, 2)$ is the mid point of PQ . The distance of the centroid of ΔPQR from the point of intersection of the line $\frac{x-2}{0} = \frac{y}{2} = \frac{z+3}{-1}$ and $\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z+1}{1}$ is 29 Jan 2024 (M)

- (1) 69 (2) 9
(3) $\sqrt{69}$ (4) $\sqrt{99}$

Q37. Let O be the origin, and M and N be the points on the lines $\frac{x-5}{4} = \frac{y-4}{1} = \frac{z-5}{3}$ and $\frac{x+8}{12} = \frac{y+2}{5} = \frac{z+11}{9}$ respectively such that MN is the shortest distance between the given lines. Then $\vec{OM} \cdot \vec{ON}$ is equal to 29 Jan 2024 (E)

_____.

Q38. A line with direction ratio $2, 1, 2$ meets the lines $x = y + 2 = z$ and $x + 2 = 2y = 2z$ respectively at the point P and Q . if the length of the perpendicular from the point $(1, 2, 12)$ to the line PQ is l , then l^2 is 29 Jan 2024 (M)

Q39. The position vectors of the vertices A , B and C of a triangle are $2\hat{i} - 3\hat{j} + 3\hat{k}$, $2\hat{i} + 2\hat{j} + 3\hat{k}$ and $-\hat{i} + \hat{j} + 3\hat{k}$ respectively. Let l denotes the length of the angle bisector AD of $\angle BAC$ where D is on the line segment BC , then $2l^2$ equals : 27 Jan 2024 (E)

- (1) 49 (2) 42
(3) 50 (4) 45

Q40. Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} + 2\hat{k}$ and $2\hat{i} + \hat{j} + 2\hat{k}$ respectively. Let l_1, l_2 and l_3 be the lengths of perpendiculars drawn from the ortho centre of the triangle on the sides AB, BC and CA respectively, then $l_1^2 + l_2^2 + l_3^2$ equals :

27 Jan 2024 (E)

- (1) $\frac{1}{5}$ (2) $\frac{1}{2}$
(3) $\frac{1}{4}$ (4) $\frac{1}{3}$

Q41. The distance, of the point $(7, -2, 11)$ from the line $\frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$ along the line $\frac{x-5}{2} = \frac{y-1}{-3} = \frac{z-5}{6}$, is :

27 Jan 2024 (M)

- (1) 12 (2) 14
(3) 18 (4) 21

Q42. If the shortest distance between the lines $\frac{x-4}{1} = \frac{y+1}{2} = \frac{z}{-3}$ and $\frac{x-\lambda}{2} = \frac{y+1}{4} = \frac{z-2}{-5}$ is $\frac{6}{\sqrt{5}}$, then the sum of all possible values of λ is :

27 Jan 2024 (M)

- (1) 5 (2) 8
(3) 7 (4) 10

Q43. The lines $\frac{x-2}{2} = \frac{y}{-2} = \frac{z-7}{16}$ and $\frac{x+3}{4} = \frac{y+2}{3} = \frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2} = \frac{y-1}{3} = \frac{z-1}{1}$ is l , then $14l^2$ is equal to _____.

27 Jan 2024 (E)

Q44. Let the image of the point $(1, 0, 7)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ be the point (α, β, γ) . Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis?

27 Jan 2024 (E)

- (1) $(1, -2, 1 + \sqrt{2})$ (2) $(1, 2, 1 - \sqrt{2})$
(3) $(3, 4, 3 - 2\sqrt{2})$ (4) $(3, -4, 3 + 2\sqrt{2})$

Q45. Let S be the set of all values of λ , for which the shortest distance between the lines $\frac{x-\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ is 13. Then $8|\sum_{\lambda \in S} \lambda|$ is equal to

15 Apr 2023 (M)

- (1) 306 (2) 304
(3) 308 (4) 302

Q46*. Let the foot of perpendicular of the point $P(3, -2, -9)$ on the plane passing through the points

$(-1, -2, -3), (9, 3, 4), (9, -2, 1)$ be $Q(\alpha, \beta, \gamma)$. Then the distance Q from the origin is

15 Apr 2023 (M)

- (1) $\sqrt{42}$ (2) $\sqrt{38}$
(3) $\sqrt{35}$ (4) $\sqrt{29}$

Q47*. Let the plane P contain the line $2x + y - z - 3 = 0 = 5x - 3y + 4z + 9$ and be parallel to the line

$\frac{x+2}{2} = \frac{3-y}{-4} = \frac{z-7}{5}$. Then the distance of the point $A(8, -1, -19)$ from the plane P measured parallel to the line $\frac{x-3}{3} = \frac{y-5}{4} = \frac{2-z}{-12}$ is equal to _____.

15 Apr 2023 (M)

Q48*. If the line $x = y = z$ intersects the line

$x \sin A + y \sin B + z \sin C - 18 = 0 = x \sin 2A + y \sin 2B + z \sin 2C - 9$, where A, B, C are the angles of a triangle ABC , then $80(\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2})$ is equal to _____.

15 Apr 2023 (M)

Q49. Let N be the foot of perpendicular from the point $P(1, -2, 3)$ on the line passing through the points $(4, 5, 8)$ and $(1, -7, 5)$. Then the distance of N from the plane $2x - 2y + z + 5 = 0$ is 13 Apr 2023 (E)

- (1) 8 (2) 6
(3) 9 (4) 7

Q50*. Let the image of the point $(\frac{5}{3}, \frac{5}{3}, \frac{8}{3})$ in the plane $x - 2y + z - 2 = 0$ be P . If the distance of the point $Q(6, -2, \alpha)$, $\alpha > 0$, from P is 13, then α is equal to _____ 13 Apr 2023 (M)

Q51*. The plane, passing through the points $(0, -1, 2)$ and $(-1, 2, 1)$ and parallel to the line passing through $(5, 1, -7)$ and $(1, -1, -1)$, also passes through the point 13 Apr 2023 (E)

- (1) $(-2, 5, 0)$ (2) $(1, -2, 1)$
(3) $(2, 0, 1)$ (4) $(0, 5, -2)$

Q52*. The line, that is coplanar to the line $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$, is 13 Apr 2023 (E)

- (1) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{4}$ (2) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$
(3) $\frac{x-1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$ (4) $\frac{x+1}{1} = \frac{y-2}{2} = \frac{z-5}{5}$

Q53*. Let the equation of plane passing through the line of intersection of the planes $x + 2y + az = 2$ and $x - y + z = 3$ be $5x - 11y + bz = 6a - 1$. For $c \in \mathbb{Z}$, if the distance of this plane from the point $(a, -c, c)$ is $\frac{2}{\sqrt{a}}$, then $\frac{a+b}{c}$ is equal to 13 Apr 2023 (M)

- (1) 2 (2) 4
(3) -4 (4) -2

Q54*. The distance of the point $(-1, 2, 3)$ from the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 10$ parallel to the line of the shortest distance between the lines $\vec{r} = (\hat{i} - \hat{j}) + \lambda(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k})$ is 13 Apr 2023 (M)

- (1) $4\sqrt{6}$ (2) $2\sqrt{5}$
(3) $2\sqrt{6}$ (4) $3\sqrt{5}$

Q55*. Let the plane $P : 4x - y + z = 10$ be rotated by an angle $\frac{\pi}{2}$ about its line of intersection with the plane $x + y - z = 4$. If α is the distance of the point $(2, 3, -4)$ from the new position of the plane P , then 35α is equal to 12 Apr 2023 (M)

- (1) 85 (2) 105
(3) 126 (4) 90

Q56*. Let the lines $L_1 : \frac{x+5}{3} = \frac{y+4}{1} = \frac{z-\alpha}{-2}$ and $L_2 : 3x + 2y + z - 2 = 0 = x - 3y + 2z - 13$ be coplanar. If the point $P(a, b, c)$ on L_1 is nearest to the point $Q(-4, -3, 2)$, then $|a| + |b| + |c|$ is equal to 12 Apr 2023 (M)

- (1) 12 (2) 14
(3) 8 (4) 10

Q57*. Let (α, β, γ) be the image of point $P(2, 3, 5)$ in the plane $2x + y - 3z = 6$. Then $\alpha + \beta + \gamma$ is equal to

11 Apr 2023 (M)

(1) 5

(2) 10

(3) 12

(4) 9

Q58*. If the equation of the plane that contains the point $(-2, 3, 5)$ and is perpendicular to each of the planes

$$2x + 4y + 5z = 8 \text{ and } 3x - 2y + 3z = 5 \text{ is } \alpha x + \beta y + \gamma z + 97 = 0 \text{ then } \alpha + \beta + \gamma =$$

11 Apr 2023 (M)

(1) 15

(2) 18

(3) 16

(4) 17

Q59*. Let the line $L : x = \frac{1-y}{2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$ meet the plane $P : x + 2y + 3z = 4$ at the point (α, β, γ) . If the angle between the line L and the plane P is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then $\alpha + 2\beta + 6\gamma$ is equal to

11 Apr 2023 (E)

Q60. Let a line L pass through the origin and be perpendicular to the lines

$$L_1 : \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R} \text{ and}$$

$$L_2 : \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}. \text{ If } P \text{ is the point of intersection of } L \text{ and } L_1, \text{ and } Q(\alpha, \beta, \gamma) \text{ is}$$

the foot of perpendicular from P on L_2 , then $9(\alpha + \beta + \gamma)$ is equal to _____.

11 Apr 2023 (M)

Q61*. Let P be the plane passing through the points $(5, 3, 0)$, $(13, 3, -2)$ and $(1, 6, 2)$. For $\alpha \in \mathbb{N}$, if the distance of the points $A(3, 4, \alpha)$ and $B(2, \alpha, a)$ from the plane P are 2 and 3 respectively, then the positive value of a is

11 Apr 2023 (E)

(1) 6

(2) 3

(3) 5

(4) 4

Q62*. Let the line passing through the points $P(2, -1, 2)$ and $Q(5, 3, 4)$ meet the plane $x - y + z = 4$ at the point R . Then the distance of the point R from the plane $x + 2y + 3z + 2 = 0$ measured parallel to the line

$$\frac{x-7}{2} = \frac{y+3}{2} = \frac{z-2}{1} \text{ is}$$

11 Apr 2023 (E)

(1) $\sqrt{61}$ (2) $\sqrt{189}$ (3) $\sqrt{31}$

(4) 3

Q63. The shortest distance between the lines $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ is

10 Apr 2023 (M)

(1) 8

(2) 6

(3) 7

(4) 9

Q64. Let the line $\frac{x}{1} = \frac{6-y}{2} = \frac{z+8}{5}$ intersect the lines $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1}$ and $\frac{x+3}{6} = \frac{3-y}{3} = \frac{z-6}{1}$ at the points A and B respectively. Then the distance of the mid-point of the line segment AB from the plane $2x - 2y + z = 14$ is

10 Apr 2023 (E)

(1) 3

(2) $\frac{11}{3}$

(3) 4

(4) $\frac{10}{3}$

Q65*. Let two vertices of a triangle ABC be $(2, 4, 6)$ and $(0, -2, -5)$, and its centroid be $(2, 1, -1)$. If the image of the third vertex in the plane $x + 2y + 4z = 11$ is (α, β, γ) , then $\alpha\beta + \beta\gamma + \gamma\alpha$ is equal to

10 Apr 2023 (M)

(1) 70

(3) 74

(2) 76

(4) 72

Q66*. Let the image of the point $P(1, 2, 6)$ in the plane passing through the points $A(1, 2, 0)$ and $B(1, 4, 1)$ $C(0, 5, 1)$ be $Q(\alpha, \beta, \gamma)$. Then $(\alpha^2 + \beta^2 + \gamma^2)$ equal to **10 Apr 2023 (E)**

(1) 65

(3) 76

(2) 62

(4) 70

Q67*. Let the foot of perpendicular from the point $A(4, 3, 1)$ on the plane $P: x - y + 2z + 3 = 0$ be N . If $B(5, \alpha, \beta)$, $\alpha, \beta \in \mathbb{Z}$ is a point on plane P such that the area of the triangle ABN is $3\sqrt{2}$, then $\alpha^2 + \beta^2 + \alpha\beta$ is equal to **10 Apr 2023 (E)**

Q68*. Let P be the point of intersection of the line $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ and the plane $x + y + z = 2$. If the distance of the point P from the plane $3x - 4y + 12z = 32$ is q , then q and $2q$ are the roots of the equation **10 Apr 2023 (M)**

(1) $x^2 - 18x - 72 = 0$ (3) $x^2 + 18x + 72 = 0$ (2) $x^2 - 18x + 72 = 0$ (4) $x^2 + 18x - 72 = 0$

Q69. The shortest distance between the lines $\frac{x-4}{4} = \frac{y+2}{5} = \frac{z+3}{3}$ and $\frac{x-1}{3} = \frac{y-3}{4} = \frac{z-4}{2}$ is **08 Apr 2023 (M)**

(1) $6\sqrt{3}$ (3) $6\sqrt{2}$ (2) $2\sqrt{6}$ (4) $3\sqrt{6}$

Q70. Let λ_1, λ_2 be the values of λ for which the points $(\frac{5}{2}, 1, \lambda)$ and $(-2, 0, 1)$ are at equal distance from the plane $2x + 3y - 6z + 7$. If $\lambda_1 > \lambda_2$ then the distance of the point $(\lambda_1 - \lambda_2, \lambda_2, \lambda_1)$ from the line $\frac{x-5}{1} = \frac{y-1}{2} = \frac{z+7}{2}$ is **08 Apr 2023 (M)**

Q71*. Let P be the plane passing through the line $\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ and the point $(2, 4, -3)$. If the image of the point $(-1, 3, 4)$ in the plane P is (α, β, γ) , then $\alpha + \beta + \gamma$ is equal to **08 Apr 2023 (E)**

(1) 10

(3) 9

(2) 12

(4) 11

Q72*. If the equation of the plane containing the line $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ and perpendicular to the plane $\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ is $ax + by + cz = 4$ then $(a - b + c)$ is equal to **08 Apr 2023 (M)**

(1) 18

(3) 20

(2) 22

(4) 24

Q73*. For $a, b \in \mathbb{Z}$ and $|a - b| \leq 10$, let the angle between the plane $P: ax + y - z = b$ and the line $L: x - 1 = a - y = z + 1$ be $\cos^{-1}(\frac{1}{3})$. If the distance of the point $(6, -6, 4)$ from the plane P is $3\sqrt{6}$, then $a^4 + b^2$ is equal to **08 Apr 2023 (E)**

(1) 32

(3) 25

(2) 85

(4) 48

Q74*. Let P_1 be the plane $3x - y - 7z = 11$ and P_2 be the plane passing through the points $(2, -1, 0)$, $(2, 0, -1)$, and $(5, 1, 1)$. If the foot of the perpendicular drawn from the point $(7, 4, -1)$ on the line of intersection of the

planes P_1 and P_2 is (α, β, γ) , then $\alpha + \beta + \gamma$ is equal to

08 Apr 2023 (E)

Q75. If the lines $\frac{x-1}{2} = \frac{2-y}{-3} = \frac{z-3}{\alpha}$ and $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{\beta}$ intersect, then the magnitude of the minimum value of $8\alpha\beta$ is _____.

06 Apr 2023 (E)

Q76*. If the equation of the plane passing through the line of intersection of the planes

$2x - y + z = 3$, $4x - 3y + 5z + 9 = 0$ and parallel to the line $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ is $ax + by + cz + 6 = 0$, then $a + b + c$ is equal to

06 Apr 2023 (M)

(1) 12

(2) 14

(3) 16

(4) 13

Q77. One vertex of a rectangular parallelopiped is at the origin O and the lengths of its edges along x , y and z axes are 3, 4 and 5 units respectively. Let P be the vertex $(3, 4, 5)$. Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is

06 Apr 2023 (M)

(1) $\frac{12}{\sqrt{5}}$

(2) $12\sqrt{5}$

(3) $\frac{12}{5\sqrt{5}}$

(4) $\frac{12}{5}$

Q78. Let the line L pass through the point $(0, 1, 2)$, intersect the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and be parallel to the plane $2x + y - 3z = 4$. Then the distance of the point $P(1, -9, 2)$ from the line L is

06 Apr 2023 (E)

(1) $\sqrt{74}$

(2) $\sqrt{69}$

(3) $\sqrt{54}$

(4) 9

Q79*. Let the image of the point $P(1, 2, 3)$ in the plane $2x - y + z = 9$ be Q . If the coordinates of the point R are $(6, 10, 7)$, then the square of the area of the triangle PQR is _____.

06 Apr 2023 (M)

Q80*. A plane P contains the line of intersection of the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -5$. If P passes through the point $(0, 2, -2)$, then the square of distance of the point $(12, 12, 18)$ from the plane P is

06 Apr 2023 (E)

(1) 620

(2) 155

(3) 310

(4) 1240

Q81. The shortest distance between the lines $\frac{x-5}{1} = \frac{y-2}{2} = \frac{z-4}{-3}$ and $\frac{x+3}{1} = \frac{y+5}{4} = \frac{z-1}{-5}$ is

01 Feb 2023 (M)

(1) $7\sqrt{3}$

(2) $5\sqrt{3}$

(3) $6\sqrt{3}$

(4) $4\sqrt{3}$

Q82*. Let $\alpha x + \beta y + \gamma z = 1$ be the equation of a plane passing through the point $(3, -2, 5)$ and perpendicular to the line joining the points $(1, 2, 3)$ and $(-2, 3, 5)$. Then the value of $\alpha \beta \gamma$ is equal to _____.

01 Feb 2023 (E)

Q83*. Let the image of the point $P(2, -1, 3)$ in the plane $x + 2y - z = 0$ be Q . Then the distance of the plane

$3x + 2y + z + 29 = 0$ from the point Q is

01 Feb 2023 (M)

(1) $\frac{22\sqrt{2}}{7}$
(3) $2\sqrt{14}$

(2) $\frac{24\sqrt{2}}{7}$
(4) $3\sqrt{14}$

Q84*. Let the plane P pass through the intersection of the planes $2x + 3y - z = 2$ and $x + 2y + 3z = 6$, and be perpendicular to the plane $2x + y - z + 1 = 0$. If d is the distance of P from the point $(-7, 1, 1)$, then d^2 is equal to :

01 Feb 2023 (E)

(1) $\frac{250}{83}$
(3) $\frac{25}{83}$

(2) $\frac{15}{53}$
(4) $\frac{250}{82}$

Q85*. The point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points $A(-3, -6, 1)$ and $B(2, 4, -3)$ divides the line segment AB internally in the ratio $k : 1$. If a, b, c ($|a|, |b|, |c|$ are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$, then $|a + b + c|$ is equal to _____.

01 Feb 2023 (E)

Q86*. Let the plane $P : 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ be parallel to the line $L : \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$. If the intercept of P on the y -axis is 1, then the distance between P and L is

31 Jan 2023 (E)

(1) $\sqrt{\frac{2}{7}}$
(3) $\sqrt{\frac{7}{2}}$

(2) $\frac{6}{\sqrt{14}}$
(4) $\sqrt{14}$

Q87*. Let P be the plane, passing through the point $(1, -1, -5)$ and perpendicular to the line joining the points $(4, 1, -3)$ and $(2, 4, 3)$. Then the distance of P from the point $(3, -2, 2)$ is

31 Jan 2023 (E)

(1) 6
(3) 5

(2) 4
(4) 7

Q88. Let the shortest distance between the lines $L : \frac{x-5}{-2} = \frac{y-\lambda}{0} = \frac{z+\lambda}{1}, \lambda \geq 0$ and $L_1 : x + 1 = y - 1 = 4 - z$ be $2\sqrt{6}$. If (α, β, γ) lies on L , then which of the following is NOT possible?

31 Jan 2023 (M)

(1) $\alpha + 2\gamma = 24$
(3) $2\alpha - \gamma = 9$

(2) $2\alpha + \gamma = 7$
(4) $\alpha - 2\gamma = 19$

Q89*. The foot of perpendicular from the origin O to a plane P which meets the co-ordinate axes at the point A, B, C is $(2, a, 4), a \in N$. If the volume of the tetrahedron $OABC$ is 144 unit^3 , then which of the following points is NOT on P ?

31 Jan 2023 (E)

(1) $(0, 4, 4)$
(3) $(0, 6, 3)$

(2) $(3, 0, 4)$
(4) $(2, 2, 4)$

Q90*. Let the line $L : \frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{1}$ intersect the plane $2x + y + 3z = 16$ at the point P . Let the point Q be the foot of perpendicular from the point $R(1, -1, -3)$ on the line L . If α is the area of triangle PQR , then α^2 is equal to _____.

31 Jan 2023 (M)

Q91*. Let θ be the angle between the planes $P_1 = \vec{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 9$ and $P_2 = \vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 15$. Let L be the line that meets P_2 at the point $(4, -2, 5)$ and makes an angle θ with the normal of P_2 . If α is the angle between L and P_2 then $(\tan^2 \theta)(\cot^2 \alpha)$ is equal to _____.

31 Jan 2023 (M)

Q92* If the equation of the plane passing through the point $(1, 1, 2)$ and perpendicular to the line $x - 3y + 2z - 1 = 0 = 4x - y + z$ is $Ax + By + Cz = 1$, then $140(C - B + A)$ is equal to 30 Jan 2023 (M)

Q93* If $\lambda_1 < \lambda_2$ are two values of λ such that the angle between the planes $P_1 : \vec{r} \cdot (3\hat{i} - 5\hat{j} + \hat{k}) = 7$ and $P_2 : \vec{r} \cdot (\lambda\hat{i} + \hat{j} - 3\hat{k}) = 9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1, 10\lambda_2, 2)$ to the plane P_1 is 30 Jan 2023 (M)

Q94* A vector $\vec{v}$ in the first octant is inclined to the x axis at 60° , to the y -axis at 45° and to the z -axis at an acute angle. If a plane passing through the points $(\sqrt{2}, -1, 1)$ and (a, b, c) , is normal to $\vec{v}$, then 30 Jan 2023 (E)

(1) $\sqrt{2}a + b + c = 1$ (2) $a + b + \sqrt{2}c = 1$
 (3) $a + \sqrt{2}b + c = 1$ (4) $\sqrt{2}a - b + c = 1$

Q95* If a plane passes through the points $(-1, k, 0)$, $(2, k, -1)$, $(1, 1, 2)$ and is parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ is 30 Jan 2023 (E)

(1) $\frac{17}{5}$ (2) $\frac{5}{17}$
 (3) $\frac{6}{13}$ (4) $\frac{13}{6}$

Q96. Let a line L pass through the point $P(2, 3, 1)$ and be parallel to the line $x + 3y - 2z - 2 = 0 = x - y + 2z$. If the distance of L from the point $(5, 3, 8)$ is α , then $3\alpha^2$ is equal to 30 Jan 2023 (E)

Q97* Let a unit vector $\vec{OP}$ make angle α, β, γ with the positive directions of the co-ordinate axes OX, OY, OZ respectively, where $\beta \in (0, \frac{\pi}{2})$. $\vec{OP}$ is perpendicular to the plane through points $(1, 2, 3)$, $(2, 3, 4)$ and $(1, 5, 7)$, then which one of the following is true? 30 Jan 2023 (M)

(1) $\alpha \in (\frac{\pi}{2}, \pi)$ and $\gamma \in (\frac{\pi}{2}, \pi)$ (2) $\alpha \in (0, \frac{\pi}{2})$ and $\gamma \in (0, \frac{\pi}{2})$
 (3) $\alpha \in (\frac{\pi}{2}, \pi)$ and $\gamma \in (0, \frac{\pi}{2})$ (4) $\alpha \in (0, \frac{\pi}{2})$ and $\gamma \in (\frac{\pi}{2}, \pi)$

Q98. The line l_1 passes through the point $(2, 6, 2)$ and is perpendicular to the plane $2x + y - 2z = 10$. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is: 30 Jan 2023 (M)

(1) 7 (2) $\frac{19}{3}$
 (3) $\frac{19}{2}$ (4) 9

Q99. Shortest distance between the lines $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ and $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ is 29 Jan 2023 (E)

(1) $2\sqrt{3}$ (2) $4\sqrt{3}$
 (3) $3\sqrt{3}$ (4) $5\sqrt{3}$

Q100. If the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ and $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ intersects at the point P , then the distance of the point P from the plane $z = a$ is: 29 Jan 2023 (E)

(1) 16 (2) 28
 (3) 10 (4) 22

Q101. Let the co-ordinates of one vertex of $\triangle ABC$ be $A(0, 2, \alpha)$ and the other two vertices lie on the line $\frac{x+\alpha}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. For $\alpha \in \mathbb{Z}$, if the area of $\triangle ABC$ is 21 sq. units and the line segment BC has length

$2\sqrt{21}$ units, then α^2 is equal to _____.

29 Jan 2023 (M)

Q102*. Let the equation of the plane P containing the line $x + 10 = \frac{8-y}{2} = z$ be $ax + by + 3z = 2(a + b)$ and the distance of the plane P from the point $(1, 27, 7)$ be c . Then $a^2 + b^2 + c^2$ is equal to

29 Jan 2023 (M)

Q103*. The plane $2x - y + z = 4$ intersects the line segment joining the points $A(a, -2, 4)$ and $B(2, b, -3)$ at the point C in the ratio $2 : 1$ and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to:

29 Jan 2023 (E)

(1) $\frac{17}{3}$
(3) $\frac{73}{3}$

(2) $\frac{16}{3}$
(4) $\frac{97}{3}$

Q104. The distance of the point $P(4, 6, -2)$ from the line passing through the point $(-3, 2, 3)$ and parallel to a line with direction ratios $(3, 3, -1)$ is equal to:

25 Jan 2023 (M)

(1) 3
(3) $2\sqrt{3}$

(2) $\sqrt{6}$
(4) $\sqrt{14}$

Q105. The shortest distance between the lines $x + 1 = 2y = -12z$ and $x = y + 2 = 6z - 6$ is

25 Jan 2023 (E)

(1) 2
(3) $\frac{5}{2}$

(2) 3
(4) $\frac{3}{2}$

Q106. Consider the lines L_1 and L_2 given by

$$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$$

$$L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$$

A line L_3 having direction ratios $1, -1, -2$, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is

25 Jan 2023 (M)

(1) $2\sqrt{6}$
(3) $4\sqrt{3}$

(2) $3\sqrt{2}$
(4) 4

Q107. The foot of perpendicular of the point $(2, 0, 5)$ on the line $\frac{x+1}{2} = \frac{y-1}{5} = \frac{z+1}{-1}$ is (α, β, γ) . Then. Which of the following is NOT correct?

25 Jan 2023 (E)

(1) $\frac{\alpha\beta}{\gamma} = \frac{4}{15}$
(3) $\frac{\beta}{\gamma} = -5$

(2) $\frac{\alpha}{\beta} = -8$
(4) $\frac{\gamma}{\alpha} = \frac{5}{8}$

Q108. If the shortest distance between the line joining the points $(1, 2, 3)$ and $(2, 3, 4)$, and the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-2}{0}$ is α , then $28\alpha^2$ is equal to _____.

25 Jan 2023 (E)

Q109*. Let the equation of the plane passing through the line

$$x - 2y - z - 5 = 0 = x + y + 3z - 5 \text{ and parallel to the line } x + y + 2z - 7 = 0 = 2x + 3y + z - 2 \text{ be}$$

$$ax + by + cz = 65. \text{ Then the distance of the point } (a, b, c) \text{ from the plane } 2x + 2y - z + 16 = 0 \text{ is } \underline{\hspace{2cm}}.$$

25 Jan 2023 (M)

Q110. The shortest distance between the lines $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ and $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ is equal to _____

24 Jan 2023 (M)

Q111* The distance of the point $(7, -3, -4)$ from the plane containing the points $(2, -3, 1)$, $(-1, 1, -2)$ and $(3, -4, 2)$ is equal to: 24 Jan 2023 (M)

- (1) 4 (2) 5
(3) $5\sqrt{2}$ (4) $4\sqrt{2}$

Q112* The distance of the point $(-1, 9, -16)$ from the plane $2x + 3y - z = 5$ measure parallel to the line $\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ is 24 Jan 2023 (M)

- (1) $13\sqrt{2}$ (2) 31
(3) 26 (4) $20\sqrt{3}$

Q113. If the shortest distance between the lines $\frac{x+\sqrt{6}}{2} = \frac{y-\sqrt{6}}{3} = \frac{z-\sqrt{6}}{4}$ and $\frac{x-\lambda}{3} = \frac{y-2\sqrt{6}}{4} = \frac{z+2\sqrt{6}}{5}$ is 6, then sum of squares of all possible values(s) of λ is 24 Jan 2023 (E)

Q114* Let the plane containing the line of intersection of the planes $P_1 : x + (\lambda + 4)y + z = 1$ and $P_2 : 2x + y + z = 2$ pass through the points $(0, 1, 0)$ and $(1, 0, 1)$. Then the distance of the point $(2\lambda, \lambda, -\lambda)$ from the plane P_2 is 24 Jan 2023 (E)

- (1) $5\sqrt{6}$ (2) $4\sqrt{6}$
(3) $2\sqrt{6}$ (4) $3\sqrt{6}$

Q115* If the foot of the perpendicular drawn from $(1, 9, 7)$ to the line passing through the point $(3, 2, 1)$ and parallel the planes $x + 2y + z = 0$ and $3y - z = 3$ is (α, β, γ) , then $\alpha + \beta + \gamma$ is equal to 24 Jan 2023 (E)

- (1) -1 (2) 3
(3) 1 (4) 5

Q116* If the foot of the perpendicular from the point $A(-1, 4, 3)$ on the plane $P : 2x + my + nz = 4$, is $(-2, \frac{7}{2}, \frac{3}{2})$, then the distance of the point A from the plane P , measured parallel to a line with direction ratios $3, -1, -4$, is equal to 29 Jul 2022 (M)

- (1) 1 (2) $\sqrt{26}$
(3) $2\sqrt{2}$ (4) $\sqrt{14}$

Q117* Let Q be the foot of perpendicular drawn from the point $P(1, 2, 3)$ to the plane $x + 2y + z = 14$. If R is a point on the plane such that $\angle PRQ = 60^\circ$, then the area of ΔPQR is equal to 29 Jul 2022 (E)

- (1) $\frac{\sqrt{3}}{2}$ (2) $\sqrt{3}$
(3) $2\sqrt{3}$ (4) 3

Q118* Let a line with direction ratios $a, -4a, -7$ be perpendicular to the lines with direction ratios $3, -1, 2b$ and $b, a, -2$. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane $x - y + z = 0$ is (α, β, γ) , then $\alpha + \beta + \gamma$ is equal to 29 Jul 2022 (M)

Q119. Let $P(-2, -1, 1)$ and $Q(\frac{56}{17}, \frac{43}{17}, \frac{111}{17})$ be the vertices of the rhombus $PRQS$. If the direction ratios of the diagonal RS are $\alpha, -1, \beta$, where both α and β are integers of minimum absolute values, then $\alpha^2 + \beta^2$ is equal to 28 Jul 2022 (M)

Q120* Let the lines $\frac{x-1}{\lambda} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x+26}{-2} = \frac{y+18}{3} = \frac{z+28}{\lambda}$ be coplanar and P be the plane containing these two lines. Then which of the following points does NOT lies on P ? 28 Jul 2022 (E)

(1) $(0, -2, -2)$

(3) $(3, -1, 0)$

(2) $(-5, 0, -1)$

(4) $(0, 4, 5)$

Q121*. The foot of the perpendicular from a point on the circle $x^2 + y^2 = 1, z = 0$ to the plane $2x + 3y + z = 6$

lies on which one of the following curves?

28 Jul 2022 (M)

(1) $(6x + 5y - 12)^2 + 4(3x + 7y - 8)^2 = 1, z = 6 - 2x - 3y$

(2) $(5x + 6y - 12)^2 + 4(3x + 5y - 9)^2 = 1, z = 6 - 2x - 3y$

(3) $(6x + 5y - 14)^2 + 9(3x + 5y - 7)^2 = 1, z = 6 - 2x - 3y$

(4) $(5x + 6y - 14)^2 + 9(3x + 7y - 8)^2 = 1, z = 6 - 2x - 3y$

Q122*. A plane P is parallel to two lines whose direction ratios are $-2, 1, -3$, and $-1, 2, -2$ and it contains the point $(2, 2, -2)$. Let P intersect the co-ordinate axes at the points A, B, C making the intercepts α, β, γ . If V is the volume of the tetrahedron $OABC$, where O is the origin and $p = \alpha + \beta + \gamma$, then the ordered pair

(V, p) is equal to

28 Jul 2022 (E)

(1) $(48, -13)$

(2) $(24, -13)$

(3) $(48, 11)$

(4) $(24, -5)$

Q123. If the length of the perpendicular drawn from the point $P(a, 4, 2), a > 0$ on the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$ is

$2\sqrt{6}$ units and $Q(\alpha_1, \alpha_2, \alpha_3)$ is the image of the point P in this line, then $a + \sum_{i=1}^3 \alpha_i$ is equal to

27 Jul 2022 (E)

(1) 7

(2) 8

(3) 12

(4) 14

Q124*. If the plane P passes through the intersection of two mutually perpendicular planes $2x + ky - 5z = 1$ and $3kx - ky + z = 5, k < 3$ and intercepts a unit length on positive x -axis, then the intercept made by the plane P on the y -axis is

27 Jul 2022 (M)

(1) $\frac{1}{11}$

(2) $\frac{5}{11}$

(3) 6

(4) 7

Q125*. If the line of intersection of the planes $ax + by = 3$ and $ax + by + cz = 0, a > 0$ makes an angle 30° with the plane $y - z + 2 = 0$, then the direction cosines of the line are

27 Jul 2022 (E)

(1) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$

(2) $\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0$

(3) $\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$

(4) $\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

Q126*. Let the line $\frac{x-3}{7} = \frac{y-2}{-1} = \frac{z-3}{-4}$ intersect the plane containing the lines $\frac{x-4}{1} = \frac{y+1}{-2} = \frac{z}{1}$ and

$4ax - y + 5z - 7a = 0 = 2x - 5y - z - 3, a \in \mathbb{R}$ at the point $P(\alpha, \beta, \gamma)$. Then the value of $\alpha + \beta + \gamma$

equals _____.

27 Jul 2022 (M)

Q127. The length of the perpendicular from the point $(1, -2, 5)$ on the line passing through $(1, 2, 4)$ and parallel to the line $x + y - z = 0 = x - 2y + 3z - 5$ is:

26 Jul 2022 (M)

(1) $\sqrt{\frac{21}{2}}$

(2) $\sqrt{\frac{9}{2}}$

(3) $\sqrt{\frac{73}{2}}$

(4) 1

Q128*. A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i} + \hat{j}$ and the plane determined by the vectors $\hat{i} - \hat{j}, \hat{i} + \hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$ is _____.

26 Jul 2022 (E)

- (1) $\frac{3\pi}{4}$ (2) $\frac{2\pi}{3}$
 (3) $\frac{4\pi}{5}$ (4) $\frac{5\pi}{6}$

Q129*. The largest value of a , for which the perpendicular distance of the plane containing the lines $\vec{r} = (\hat{i} + \hat{j}) + \lambda(\hat{i} + a\hat{j} - \hat{k})$ and $\vec{r} = (\hat{i} + \hat{j}) + \mu(-\hat{i} + \hat{j} - a\hat{k})$ from the point $(2, 1, 4)$ is $\sqrt{3}$, is _____.

26 Jul 2022 (E)

Q130*. The plane passing through the line $L : lx - y + 3(1 - l)z = 1, x + 2y - z = 2$ and perpendicular to the plane $3x + 2y + z = 6$ is $3x - 8y + 7z = 4$. If θ is the acute angle between the line L and the y -axis, then $415 \cos^2 \theta$ is equal to _____.

26 Jul 2022 (E)

Q131. Let Q and R be two points on the line $\frac{x+1}{2} = \frac{y+2}{3} = \frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point $P(4, 2, 7)$. Then the square of the area of the triangle PQR is _____.

26 Jul 2022 (M)

Q132. The shortest distance between the lines $\frac{x+7}{-6} = \frac{y-6}{7} = z$ and $\frac{7-x}{2} = y - 2 = z - 6$ is _____.

25 Jul 2022 (E)

- (1) $2\sqrt{29}$ (2) 1
 (3) $\sqrt{\frac{37}{29}}$ (4) $\frac{\sqrt{29}}{2}$

Q133*. A plane E is perpendicular to the two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$, and passes through the point $P(1, -1, 1)$. If the distance of the plane E from the point $Q(a, a, 2)$ is $3\sqrt{2}$, then $(PQ)^2$ is equal to _____.

25 Jul 2022 (E)

- (1) 9 (2) 12
 (3) 21 (4) 33

Q134*. Let P be the plane containing the straight line $\frac{x-3}{9} = \frac{y+4}{-1} = \frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2} = \frac{y}{3} = \frac{z}{5}$ and $\frac{x}{3} = \frac{y}{7} = \frac{z}{8}$. If d is the distance of P from the point $(2, -5, 11)$, then d^2 is equal to _____.

25 Jul 2022 (M)

- (1) $\frac{147}{2}$ (2) 96
 (3) $\frac{32}{3}$ (4) 54

Q135. The line of shortest distance between the lines $\frac{x-2}{0} = \frac{y-1}{1} = \frac{z}{1}$ and $\frac{x-3}{2} = \frac{y-5}{2} = \frac{z-1}{1}$ makes an angle of $\sin^{-1}\left(\sqrt{\frac{2}{27}}\right)$ with the plane $P : ax - y - z = 0, (a > 0)$. If the image of the point $(1, 1, -5)$ in the plane P is (α, β, γ) , then $\alpha + \beta - \gamma$ is equal to _____.

25 Jul 2022 (M)

Q136*. If the mirror image of the point $(2, 4, 7)$ in the plane $3x - y + 4z = 2$ is (a, b, c) , the $2a + b + 2c$ is equal to _____.

29 Jun 2022 (M)

- (1) 54 (2) -6
 (3) 50 (4) -42

Q137*. Let d be the distance between the foot of perpendiculars of the points $P(1, 2, -1)$ and $Q(2, -1, 3)$ on the plane $-x + y + z = 1$. Then d^2 is equal to _____.

29 Jun 2022 (M)

Q138*. Let $P_1 : \vec{r} \cdot (2\hat{i} + \hat{j} - 3\hat{k}) = 4$ be a plane. Let P_2 be another plane which passes through the points $(2, -3, 2)$, $(2, -2, -3)$ and $(1, -4, 2)$. If the direction ratios of the line of intersection of P_1 and P_2 be $16\alpha, \beta$, then the value of $\alpha + \beta$ is equal to _____. **29 Jun 2022 (M)**

Q139*. Let Q be the mirror image of the point $P(1, 2, 1)$ with respect to the plane $x + 2y + 2z = 16$. Let T be a plane passing through the point Q and contains the line $\vec{r} = -\hat{k} + \lambda(\hat{i} + \hat{j} + 2\hat{k})$, $\lambda \in R$. Then, which of the following points lies on T ? **29 Jun 2022 (E)**

- (1) $(2, 1, 0)$ (2) $(1, 2, 1)$
(3) $(1, 2, 2)$ (4) $(1, 3, 2)$

Q140*. Let $\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z+3}{-1}$ lie on the plane $px - qy + z = 5$, for some $p, q \in R$. The shortest distance of the plane from the origin is: **29 Jun 2022 (E)**

- (1) $\sqrt{\frac{3}{109}}$ (2) $\sqrt{\frac{5}{142}}$
(3) $\sqrt{\frac{5}{71}}$ (4) $\sqrt{\frac{1}{142}}$

Q141. Let the image of the point $P(1, 2, 3)$ in the line $L : \frac{x-6}{3} = \frac{y-1}{2} = \frac{z-2}{3}$ be Q . Let $R(\alpha, \beta, \gamma)$ be a point that divides internally the line segment PQ in the ratio $1 : 3$. Then the value of $22(\alpha + \beta + \gamma)$ is equal to **28 Jun 2022 (E)**

Q142*. If two distinct point Q, R lie on the line of intersection of the planes $-x + 2y - z = 0$ and $3x - 5y + 2z = 0$ and $PQ = PR = \sqrt{18}$ where the point P is $(1, -2, 3)$, then the area of the triangle PQR is equal to **28 Jun 2022 (M)**

- (1) $\frac{2}{3}\sqrt{38}$ (2) $\frac{4}{3}\sqrt{38}$
(3) $\frac{8}{3}\sqrt{38}$ (4) $\sqrt{\frac{152}{3}}$

Q143*. The acute angle between the planes P_1 and P_2 , when P_1 and P_2 are the planes passing through the intersection of the planes $5x + 8y + 13z - 29 = 0$ and $8x - 7y + z - 20 = 0$ and the points $(2, 1, 3)$ and $(0, 1, 2)$, respectively, is **28 Jun 2022 (M)**

- (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{4}$
(3) $\frac{\pi}{6}$ (4) $\frac{\pi}{12}$

Q144*. Let the plane $ax + by + cz = d$ pass through $(2, 3, -5)$ and is perpendicular to the planes $2x + y - 5z = 10$ and $3x + 5y - 7z = 12$

If a, b, c, d are integers $d > 0$ and $\gcd(|a|, |b|, |c|, d) = 1$ then the value of $a + 7b + c + 20d$ is equal to **28 Jun 2022 (E)**

- (1) 18 (2) 20
(3) 24 (4) 22

Q145*. Let the plane $P : \vec{r} \cdot \vec{a} = d$ contain the line of intersection of two planes $\vec{r} \cdot (\hat{i} + 3\hat{j} - \hat{k}) = 6$ and $\vec{r} \cdot (-6\hat{i} + 5\hat{j} - \hat{k}) = 7$. If the plane P passes through the point $(2, 3, \frac{1}{2})$, then the value of $\frac{|13\vec{a}|^2}{d^2}$ is equal to **28 Jun 2022 (M)**

(1) 90

(3) 95

(2) 93

(4) 97

Q146. Let the foot of the perpendicular from the point $(1, 2, 4)$ on the line $\frac{x+2}{4} = \frac{y-1}{2} = \frac{z+1}{3}$ be P . Then the distance of P from the plane $3x + 4y + 12z + 23 = 0$ is 27 Jun 2022 (E)

(1) $\frac{50}{13}$ (3) $\frac{65}{13}$ (2) $\frac{63}{13}$

(4) 4

Q147. The shortest distance between the lines $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{-1}$ and $\frac{x+3}{2} = \frac{y-6}{1} = \frac{z-5}{3}$ is 27 Jun 2022 (E)

(1) $\frac{18}{\sqrt{5}}$ (3) $\frac{46}{3\sqrt{5}}$ (2) $\frac{22}{3\sqrt{5}}$ (4) $6\sqrt{3}$

Q148. If two straight lines whose direction cosines are given by the relations $l + m - n = 0$, $3l^2 + m^2 + cnl = 0$ are parallel, then the positive value of c is 27 Jun 2022 (M)

(1) 6

(3) 3

(2) 4

(4) 2

Q149*. Let the mirror image of the point (a, b, c) with respect to the plane $3x - 4y + 12z + 19 = 0$ be $(a - 6, \beta, \gamma)$.

If $a + b + c = 5$, then $7\beta - 9\gamma$ is equal to _____.

27 Jun 2022 (M)

Q150. If the two lines $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ are perpendicular, then an angle between the lines l_2 and $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ is 26 Jun 2022 (M)

(1) $\cos^{-1}\left(\frac{29}{4}\right)$ (3) $\cos^{-1}\left(\frac{2}{29}\right)$ (2) $\sec^{-1}\left(\frac{29}{4}\right)$ (4) $\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$

Q151*. If the plane $2x + y - 5z = 0$ is rotated about its line of intersection with the plane $3x - y + 4z - 7 = 0$ by an angle of $\frac{\pi}{2}$, then the plane after the rotation passes through the point 26 Jun 2022 (E)

(1) $(2, -2, 0)$ (3) $(1, 0, 2)$ (2) $(-2, 2, 0)$ (4) $(-1, 0, -2)$

Q152*. If the lines $\vec{r} = (\hat{i} - \hat{j} + \hat{k}) + \lambda(3\hat{j} - \hat{k})$ and $\vec{r} = (\alpha\hat{i} - \hat{j}) + \mu(2\hat{i} - 3\hat{k})$ are co-planar, the the distance of the plane containing these two lines from the point $(\alpha, 0, 0)$ is 26 Jun 2022 (E)

(1) $\frac{2}{9}$ (3) $\frac{4}{11}$ (2) $\frac{2}{11}$

(4) 2

Q153*. Let the plane $2x + 3y + z + 20 = 0$ be rotated through a right angle about its line of intersection with the plane $x - 3y + 5z = 8$. If the mirror image of the point $(2, -\frac{1}{2}, 2)$ in the rotated plane is $B(a, b, c)$, then 26 Jun 2022 (M)

(1) $\frac{a}{8} = \frac{b}{5} = \frac{c}{-4}$ (3) $\frac{a}{8} = \frac{b}{-5} = \frac{c}{4}$ (2) $\frac{a}{4} = \frac{b}{5} = \frac{c}{-2}$ (4) $\frac{a}{4} = \frac{b}{5} = \frac{c}{2}$

Q154. Let l_1 be the line in xy -plane with x and y intercepts $\frac{1}{8}$ and $\frac{1}{4\sqrt{2}}$ respectively, and l_2 be the line in zx -plane with x and z intercepts $-\frac{1}{8}$ and $-\frac{1}{6\sqrt{3}}$ respectively. If d is the shortest distance between the line l_1 and l_2 , then d^{-2} is equal to 25 Jun 2022 (E)

Q155*. Let the lines $L_1: \vec{r} = \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$, $\lambda \in \mathbf{R}$ and $L_2: \vec{r} = (\hat{i} + 3\hat{j} + \hat{k}) + \mu(\hat{i} + \hat{j} + 5\hat{k})$; $\mu \in \mathbf{R}$, intersect at the point S . If a plane $ax + by - z + d = 0$ passes through S and is parallel to the lines L_1 and L_2 , then the value of $a + b + d$ is equal to _____.

25 Jun 2022 (M)

Q156*. Let Q be the mirror image of the point $P(1, 0, 1)$ with respect to the plane $S: x + y + z = 5$. If a line L passing through $(1, -1, -1)$, parallel to the line PQ meets the plane S at R , then QR^2 is equal to

25 Jun 2022 (M)

- (1) 2 (2) 5
(3) 7 (4) 11

Q157*. Let P be the plane passing through the intersection of the planes $\vec{r} \cdot (\hat{i} + 3\hat{j} - \hat{k}) = 5$ and $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 3$, and the point $(2, 1, -2)$. Let the position vectors of the points X and Y be $\hat{i} - 2\hat{j} + 4\hat{k}$ and $5\hat{i} - \hat{j} + 2\hat{k}$ respectively. Then the points

25 Jun 2022 (E)

- (1) X and $X + Y$ are on the same side of P (2) Y and $Y - X$ are on the opposite sides of P
(3) X and Y are on the opposite sides of P (4) $X + Y$ and $X - Y$ are on the same side of P

Q158. If the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$, then the sum of all possible values of λ is:

24 Jun 2022 (E)

- (1) 16 (2) 6
(3) 12 (4) 15

Q159. Let a line having direction ratios $1, -4, 2$ intersect the lines $\frac{x-7}{3} = \frac{y-1}{-1} = \frac{z+2}{1}$ and $\frac{x}{2} = \frac{y-7}{3} = \frac{z}{1}$ at the points A and B . Then $(AB)^2$ is equal to

24 Jun 2022 (M)

Q160. If the shortest distance between the lines $\vec{r} = (-\hat{i} + 3\hat{k}) + \lambda(\hat{i} - a\hat{j})$ and $\vec{r} = (-\hat{j} + 2\hat{k}) + \mu(\hat{i} - \hat{j} + \hat{k})$ is $\sqrt{\frac{2}{3}}$, then the integral value of a is equal to _____

24 Jun 2022 (M)

Q161*. Let the points on the plane P be equidistant from the points $(-4, 2, 1)$ and $(2, -2, 3)$. Then the acute angle between the plane P and the plane $2x + y + 3z = 1$ is

24 Jun 2022 (E)

- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$
(3) $\frac{\pi}{3}$ (4) $\frac{5\pi}{12}$

Q162. The distance of line $3y - 2z - 1 = 0 = 3x - z + 4$ from the point $(2, -1, 6)$ is :

01 Sep 2021 (E)

- (1) $2\sqrt{5}$ (2) $2\sqrt{6}$
(3) $\sqrt{26}$ (4) $4\sqrt{2}$

Q163*. Let the acute angle bisector of the two planes $x - 2y - 2z + 1 = 0$ and $2x - 3y - 6z + 1 = 0$ be the plane P . Then which of the following points lies on P ?

01 Sep 2021 (E)

- (1) $(0, 2, -4)$ (2) $(-2, 0, -\frac{1}{2})$
(3) $(4, 0, -2)$ (4) $(3, 1, -\frac{1}{2})$

Q164*. Let the equation of the plane, that passes through the point $(1, 4, -3)$ and contains the line of intersection of the planes $3x - 2y + 4z - 7 = 0$ and $x + 5y - 2z + 9 = 0$, be $\alpha x + \beta y + \gamma z + 3 = 0$, then $\alpha + \beta + \gamma$ is

equal to :

(1) -15

(2) 15

(3) -23

(4) 23

31 Aug 2021 (M)

Q165*. The distance of the point $(-1, 2, -2)$ from the line of intersection of the planes $2x + 3y + 2z = 0$ and $x - 2y + z = 0$ is :

31 Aug 2021 (E)

(1) $\frac{1}{\sqrt{2}}$

(2) $\frac{\sqrt{42}}{2}$

(3) $\frac{5}{2}$

(4) $\frac{\sqrt{34}}{2}$

Q166*. The square of the distance of the point of intersection of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point $(-1, -1, 2)$ is

31 Aug 2021 (M)

Q167*. Suppose the line $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ lies on the plane $x + 3y - 2z + \beta = 0$. Then $(\alpha + \beta)$ is equal to _____

31 Aug 2021 (E)

Q168*. The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are $2, 3, -6$, is

27 Aug 2021 (M)

(1) 2

(2) 5

(3) 3

(4) 1

Q169. The angle between the straight lines, whose direction cosines l, m, n are given by the equations

$2l + 2m - n = 0$ and $mn + nl + lm = 0$, is:

27 Aug 2021 (E)

(1) $\frac{\pi}{3}$

(2) $\frac{\pi}{2}$

(3) $\cos^{-1}\left(\frac{8}{9}\right)$

(4) $\pi - \cos^{-1}\left(\frac{4}{9}\right)$

Q170*. Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ units from the origin, which contains the line of intersection of the planes $x - y - z - 1 = 0$ and $2x + y - 3z + 4 = 0$, is

27 Aug 2021 (M)

(1) $-x + 2y + 2z - 3 = 0$

(2) $3x - 4z + 3 = 0$

(3) $3x - 1y - 5z + 2 = 0$

(4) $4x - y - 5z + 2 = 0$

Q171*. The equation of the plane passing through the line of intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) + 4 = 0$ and parallel to the x -axis, is

27 Aug 2021 (E)

(1) $\vec{r} \cdot (\hat{i} + 3\hat{k}) + 6 = 0$

(2) $\vec{r} \cdot (\hat{i} - 3\hat{k}) + 6 = 0$

(3) $\vec{r} \cdot (\hat{j} - 3\hat{k}) - 6 = 0$

(4) $\vec{r} \cdot (\hat{j} - 3\hat{k}) + 6 = 0$

Q172*. Let S be the mirror image of the point $Q(1, 3, 4)$ with respect to the plane $2x - y + z + 3 = 0$ and let

$R(3, 5, \gamma)$ be a point of this plane. Then the square of the length of the line segment SR is

27 Aug 2021 (E)

Q173*. Let P be the plane passing through the point $(1, 2, 3)$ and the line of intersection of the planes

$$\vec{r} \cdot (\hat{i} + \hat{j} + 4\hat{k}) = 16 \text{ \& } \vec{r} \cdot (-\hat{i} + \hat{j} + \hat{k}) = 6.$$
 Then which of the following points does NOT lie on P ?

26 Aug 2021 (E)

(1) $(4, 2, 2)$

(2) $(6, -6, 2)$

(3) $(-8, 8, 6)$

(4) $(3, 3, 2)$

Q174*. A plane P contains the line $x + 2y + 3z + 1 = 0 = x - y - z - 6$, and is perpendicular to the plane $-2x + y + z + 8 = 0$. Then which of the following points lies on P ? 26 Aug 2021 (M)

(1) $(2, -1, 1)$

(2) $(0, 1, 1)$

(3) $(-1, 1, 2)$

(4) $(1, 0, 1)$

Q175*. Let Q be the foot of the perpendicular from the point $P(7, -2, 13)$ on the plane containing the lines

$$\frac{x+1}{6} = \frac{y-1}{7} = \frac{z-3}{8} \text{ and } \frac{x-1}{3} = \frac{y-2}{5} = \frac{z-3}{7}$$

Then $(PQ)^2$, is equal to _____.

26 Aug 2021 (E)

Q176*. Let the line L be the projection of the line $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$ in the plane $x - 2y - z = 3$. If d is the distance of the point $(0, 0, 6)$ from L , then d^2 is equal to 26 Aug 2021 (M)

Q177. For real numbers α and $\beta \neq 0$, if the point of intersection of the straight lines $\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3}$ and $\frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3}$ lies on the plane $x + 2y - z = 8$, then $\alpha - \beta$ is equal to : 27 Jul 2021 (E)

(1) 5

(2) 9

(3) 3

(4) 7

Q178. The distance of the point $P(3, 4, 4)$ from the point of intersection of the line joining the points $Q(3, -4, -5)$ and $R(2, -3, 1)$ and the plane $2x + y + z = 7$, is equal to _____. 27 Jul 2021 (E)

Q179*. Let the plane passing through the point $(-1, 0, -2)$ and perpendicular to each of the planes $2x + y - z = 2$ and $x - y - z = 3$ be $ax + by + cz + 8 = 0$. Then the value of $a + b + c$ is equal to: 27 Jul 2021 (M)

(1) 3

(2) 8

(3) 5

(4) 4

Q180*. Let a plane P pass through the point $(3, 7, -7)$ and contain the line, $\frac{x-2}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$. If distance of the plane P from the origin is d , then d^2 is equal to 27 Jul 2021 (M)

Q181*. If the lines $\frac{x-k}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x+1}{3} = \frac{y+2}{2} = \frac{z+3}{1}$ are co-planar, then the value of k is _____.

25 Jul 2021 (E)

Q182*. Let the foot of perpendicular from a point $P(1, 2, -1)$ to the straight line $L : \frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$ be N . Let a line be drawn from P parallel to the plane $x + y + 2z = 0$ which meets L at point Q . If α is the acute angle between the lines PN and PQ , then $\cos \alpha$ is equal to _____.

25 Jul 2021 (M)

(1) $\frac{1}{\sqrt{5}}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{\sqrt{3}}$

(4) $\frac{1}{2\sqrt{3}}$

Q183. If the shortest distance between the straight lines $3(x - 1) = 6(y - 2) = 2(z - 1)$ and $4(x - 2) = 2(y - \lambda) = (z - 3)$, $\lambda \in R$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to: 22 Jul 2021 (M)

(1) 3

(2) 2

(3) 5

(4) -1

Q184*. Let L be the line of intersection of planes $\vec{r} \cdot (\hat{i} - \hat{j} + 2\hat{k}) = 2$ and $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) = 2$. If $P(\alpha, \beta, \gamma)$ is the foot of perpendicular on L from the point $(1, 2, 0)$, then the value of $35(\alpha + \beta + \gamma)$ is equal to: 22 Jul 2021 (M)

(1) 101

(2) 119

(3) 143

(4) 134

Q185. If the shortest distance between the lines $\vec{r}_1 = \alpha\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$, $\lambda \in R$, $\alpha > 0$ and

$\vec{r}_2 = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$, $\mu \in R$ is 9, then α is equal to _____. **20 Jul 2021 (M)**

Q186. The lines $x = ay - 1 = z - 2$ and $x = 3y - 2 = bz - 2$, ($ab \neq 0$) are coplanar, if:

20 Jul 2021 (E)(1) $b = 1$, $a \in R - \{0\}$ (2) $a = 1$, $b \in R - \{0\}$ (3) $a = 2$, $b = 2$ (4) $a = 2$, $b = 3$

Q187*. Consider the line L given by the equation $\frac{x-3}{2} = \frac{y-1}{1} = \frac{z-2}{1}$. Let Q be the mirror image of the point

$(2, 3, -1)$ with respect to L . Let a plane P be such that it passes through Q , and the line L is perpendicular to P . Then which of the following points is on the plane P ? **20 Jul 2021 (E)**

(1) $(-1, 1, 2)$ (2) $(1, 1, 1)$ (3) $(1, 1, 2)$ (4) $(1, 2, 2)$

Q188*. Let P be a plane passing through the points $(1, 0, 1)$, $(1, -2, 1)$ and $(0, 1, -2)$. Let a vector

$\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ be such that $\vec{a}$ is parallel to the plane P , perpendicular to $(\hat{i} + 2\hat{j} + 3\hat{k})$ and

$\vec{a} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 2$, then $(\alpha - \beta + \gamma)^2$ equals _____. **20 Jul 2021 (M)**

Q189*. Let the plane $ax + by + cz + d = 0$ bisect the line joining the points $(4, -3, 1)$ and $(2, 3, -5)$ at the right angles. If a, b, c, d are integers, then the minimum value of $(a^2 + b^2 + c^2 + d^2)$ is **18 Mar 2021 (M)**

Q190*. The equation of the planes parallel to the plane $x - 2y + 2z - 3 = 0$ which are at unit distance from the point $(1, 2, 3)$ is $ax + by + cz + d = 0$. If $(b - d) = K(c - a)$, then the positive value of K is **18 Mar 2021 (M)**

Q191*. Let the mirror image of the point $(1, 3, a)$ with respect to the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) - b = 0$ be $(-3, 5, 2)$.

Then the value of $|a + b|$ is equal to _____. **18 Mar 2021 (E)**

Q192*. Let P be a plane containing the line $\frac{x-1}{3} = \frac{y+6}{4} = \frac{z+5}{2}$ and parallel to the line $\frac{x-3}{4} = \frac{y-2}{-3} = \frac{z+5}{7}$. If the point $(1, -1, \alpha)$ lies on the plane P , then the value of $|5\alpha|$ is equal to _____. **18 Mar 2021 (E)**

Q193*. The equation of the plane which contains the y -axis and passes through the point $(1, 2, 3)$ is: **17 Mar 2021 (M)**

(1) $x + 3z = 10$ (2) $x + 3z = 0$ (3) $3x + z = 6$ (4) $3x - z = 0$

Q194*. If the equation of the plane passing through the line of intersection of the planes

$2x - 7y + 4z - 3 = 0$, $3x - 5y + 4z + 11 = 0$ and the point $(-2, 1, 3)$ is $ax + by + cz - 7 = 0$, then the value of $2a + b + c - 7$ is _____. **17 Mar 2021 (M)**

Q195*. Let P be an arbitrary point having sum of the squares of the distance from the planes

$x + y + z = 0$, $lx - nz = 0$ and $x - 2y + z = 0$ equal to 9 units. If the locus of the point P is

$x^2 + y^2 + z^2 = 9$, then the value of $l - n$ is equal to _____. **17 Mar 2021 (E)**

Q196*. If the equation of plane passing through the mirror image of a point $(2, 3, 1)$ with respect to line $\frac{x+1}{2} = \frac{y-3}{1} = \frac{z+2}{-1}$ and containing the line $\frac{x-2}{3} = \frac{1-y}{2} = \frac{z+1}{1}$ is $\alpha x + \beta y + \gamma z = 24$ then $\alpha + \beta + \gamma$ is equal to: 17 Mar 2021 (E)

(1) 20

(2) 19

(3) 18

(4) 21

Q197. If the foot of the perpendicular from point $(4, 3, 8)$ on the line $L_1 : \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}$, $l \neq 0$ is $(3, 5, 7)$, then the shortest distance between the line L_1 and line $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to

16 Mar 2021 (E)(1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{6}}$ (3) $\sqrt{\frac{2}{3}}$ (4) $\frac{1}{\sqrt{3}}$

Q198*. If for $a > 0$, the feet of perpendiculars from the points $A(a, -2a, 3)$ and $B(0, 4, 5)$ on the plane $lx + my + nz = 0$ are points $C(0, -a, -1)$ and D respectively, then the length of line segment CD is equal to : 16 Mar 2021 (M)

(1) $\sqrt{31}$ (2) $\sqrt{41}$ (3) $\sqrt{55}$ (4) $\sqrt{66}$

Q199*. If the distance of the point $(1, -2, 3)$ from the plane $x + 2y - 3z + 10 = 0$ measured parallel to the line, $\frac{x-1}{3} = \frac{2-y}{m} = \frac{z+3}{1}$ is $\sqrt{\frac{7}{2}}$, then the value of $|m|$ is equal to _____. 16 Mar 2021 (E)

Q200. Let the position vectors of two points P and Q be $3\hat{i} - \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$, respectively. Let R and S be two points such that the direction ratios of lines PR and QS are $(4, -1, 2)$ and $(-2, 1, -2)$, respectively. Let lines PR and QS intersect at T . If the vector $\vec{TA}$ is perpendicular to both $\vec{PR}$ and $\vec{QS}$ and the length of vector $\vec{TA}$ is $\sqrt{5}$ units, then the modulus of a position vector of A is : 16 Mar 2021 (M)

(1) $\sqrt{482}$ (2) $\sqrt{171}$ (3) $\sqrt{5}$ (4) $\sqrt{227}$

Q201*. If (x, y, z) be an arbitrary point lying on a plane P which passes through the point $(42, 0, 0)$, $(0, 42, 0)$ and $(0, 0, 42)$, then the value of expression

$$3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)}$$

16 Mar 2021 (E)

(1) 0

(2) 3

(3) 39

(4) -45

Q202*. Let P be a plane $lx + my + nz = 0$ containing the line, $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$. If plane P divides the line segment AB joining points $A(-3, -6, 1)$ and $B(2, 4, -3)$ in ratio $k : 1$ then the value of k is equal to : 16 Mar 2021 (M)

(1) 1.5

(2) 3

(3) 2

(4) 4

Q203*. Consider the three planes

$$P_1 : 3x + 15y + 21z = 9$$

$$P_2 : x - 3y - z = 5, \text{ and}$$

$$P_3 : 2x + 10y + 14z = 5$$

Then, which one of the following is true?

26 Feb 2021 (M)

(1) P_2 and P_3 are parallel.

(2) P_1, P_2 and P_3 all are parallel.

(3) P_1 and P_2 are parallel.

(4) P_1 and P_3 are parallel.

Q204*. Let $(\lambda, 2, 1)$ be a point on the plane which passes through the point $(4, -2, 2)$. If the plane is perpendicular to the line joining the points $(-2, -21, 29)$ and $(-1, -16, 23)$, then $\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$ is equal to _____.

26 Feb 2021 (M)

Q205*. If the mirror image of the point $(1, 3, 5)$ with respect to the plane $4x - 5y + 2z = 8$ is (α, β, γ) , then $5(\alpha + \beta + \gamma)$ equals :

26 Feb 2021 (E)

(1) 43

(2) 47

(3) 41

(4) 39

Q206*. Let L be a line obtained from the intersection of two planes $x + 2y + z = 6$ and $y + 2z = 4$. If point $P(\alpha, \beta, \gamma)$ is the foot of perpendicular from $(3, 2, 1)$ on L , then the value of $21(\alpha + \beta + \gamma)$ equals:

26 Feb 2021 (E)

(1) 102

(2) 142

(3) 68

(4) 136

Q207. The equation of the line through the point $(0, 1, 2)$ and perpendicular to the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{-2}$ is :

25 Feb 2021 (M)

(1) $\frac{x}{3} = \frac{y-1}{-4} = \frac{z-2}{3}$

(2) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{3}$

(3) $\frac{x}{-3} = \frac{y-1}{4} = \frac{z-2}{3}$

(4) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{-3}$

Q208. A line l passing through origin is perpendicular to the lines

$$l_1 : \vec{r} = (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}$$

$$l_2 : \vec{r} = (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}$$

If the co-ordinates of the point in the first octant on l_2 at a distance of $\sqrt{17}$ from the point of intersection of l and l_1 are (a, b, c) , then $18(a+b+c)$ is equal to ____.

25 Feb 2021 (E)

Q209. Let α be the angle between the lines whose direction cosines satisfy the equations $l + m - n = 0$ and $l^2 + m^2 - n^2 = 0$. Then the value of $\sin^4 \alpha + \cos^4 \alpha$ is :

25 Feb 2021 (M)

(1) $\frac{5}{8}$

(2) $\frac{1}{2}$

(3) $\frac{3}{8}$

(4) $\frac{3}{4}$

Q210*. A plane passes through the points $A(1, 2, 3), B(2, 3, 1)$ and $C(2, 4, 2)$. If O is the origin and P is $(2, -1, 1)$, then the projection of $\vec{OP}$ on this plane is of length:

25 Feb 2021 (E)

(1) $\sqrt{\frac{2}{5}}$

(2) $\sqrt{\frac{2}{7}}$

(3) $\sqrt{\frac{2}{3}}$

(4) $\sqrt{\frac{2}{11}}$

Q211. Let $a, b \in \mathbb{R}$. If the mirror image of the point $P(a, 6, 9)$ with respect to the line $\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is $(20, b, -a-9)$, then $|a+b|$ is equal to: 24 Feb 2021 (E)

- (1) 86 (2) 90
(3) 84 (4) 88

Q212*. The equation of the plane passing through the point $(1, 2, -3)$ and perpendicular to the planes $3x + y - 2z = 5$ and $2x - 5y - z = 7$, is 24 Feb 2021 (M)

- (1) $11x + y + 17z + 38 = 0$ (2) $3x - 10y - 2z + 11 = 0$
(3) $6x - 5y + 2z + 10 = 0$ (4) $6x - 5y - 2z - 2 = 0$

Q213*. The distance of the point $(1, 1, 9)$ from the point of intersection of the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and the plane $x + y + z = 17$ is: 24 Feb 2021 (M)

- (1) $19\sqrt{2}$ (2) $2\sqrt{19}$
(3) $\sqrt{38}$ (4) 38

Q214. Let λ be an integer. If the shortest distance between the lines $x - \lambda = 2y - 1 = -2z$ and $x = y + 2\lambda = z - \lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is: 24 Feb 2021 (E)

Q215*. The vector equation of the plane passing through the intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (\hat{i} - 2\hat{j}) = -2$, and the point $(1, 0, 2)$ is: 24 Feb 2021 (E)

- (1) $\vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = 7$ (2) $\vec{r} \cdot (\hat{i} - 7\hat{j} + 3\hat{k}) = \frac{7}{3}$
(3) $\vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = \frac{7}{3}$ (4) $\vec{r} \cdot (3\hat{i} + 7\hat{j} + 3\hat{k}) = 7$

Q216*. The shortest distance between the lines $\frac{x-1}{0} = \frac{y+1}{-1} = \frac{z}{1}$ and $x + y + z + 1 = 0, 2x - y + z + 3 = 0$ is 06 Sep 2020 (M)

- (1) 1 (2) $\frac{1}{\sqrt{3}}$
(3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{2}$

Q217*. A plane P meets the coordinate axes at A, B and C respectively. The centroid of ΔABC is given to be $(1, 1, 2)$. Then the equation of the line through this centroid and perpendicular to the plane P is: 06 Sep 2020 (E)

- (1) $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-2}{1}$ (2) $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-2}{2}$
(3) $\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{1}$ (4) $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$

Q218. If (a, b, c) is the image of the point $(1, 2, -3)$ in the line, $\frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$, then $a + b + c$ is equal to: 05 Sep 2020 (M)

- (1) 2 (2) -1
(3) 3 (4) 1

Q219*. If for some, $\alpha \in \mathbb{R}$, the lines $L_1 : \frac{x+1}{2} = \frac{y-2}{-1} = \frac{z-1}{1}$ and $L_2 : \frac{x+2}{\alpha} = \frac{y+1}{5-\alpha} = \frac{z+1}{1}$ are coplanar, then the line L_2 passes through the point: 05 Sep 2020 (E)

- (1) $(10, 2, 2)$ (2) $(2, -10, -2)$
(3) $(10, -2, -2)$ (4) $(-2, 10, 2)$

Q220*. If the equation of a plane P , passing through the intersection of the planes, $x + 4y - z + 7 = 0$ and $3x + y + 5z = 8$ is $ax + by + 6z = 15$ for some $a, b \in R$, then the distance of the point $(3, 2, -1)$ from the plane P is **04 Sep 2020 (M)**

Q221*. The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is : **04 Sep 2020 (E)**

(1) $\frac{7}{5}$ (2) 1
(3) $\frac{1}{7}$ (4) 7

Q222. The lines $\vec{r} = (\hat{i} - \hat{j}) + l(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + m(\hat{i} + \hat{j} - \hat{k})$ **03 Sep 2020 (M)**

(1) Do not intersect for any values of l and m (2) Intersect for all values of l and m
(3) Intersect when $l = 2$ and $m = \frac{1}{2}$ (4) Intersect when $l = 1$ and $m = 2$

Q223*. The foot of the perpendicular drawn from the point $(4, 2, 3)$ to the line joining the points $(1, -2, 3)$ and $(1, 1, 0)$ lies on the plane **03 Sep 2020 (M)**

(1) $2x + y - z = 1$ (2) $x - y - 2z = 1$
(3) $x - 2y + z = 1$ (4) $x + 2y - z = 1$

Q224*. The plane which bisects the line joining the points $(4, -2, 3)$ and $(2, 4, -1)$ at right angles also passes through the point : **03 Sep 2020 (E)**

(1) $(0, -1, 1)$ (2) $(4, 0, -1)$
(3) $(4, 0, 1)$ (4) $(0, 1, -1)$

Q225*. Let a plane P contain two lines $\vec{r} = \hat{i} + \lambda(\hat{i} + \hat{j})$, $\lambda \in R$ and $\vec{r} = -\hat{j} + \mu(\hat{j} - \hat{k})$, $\mu \in R$. If $Q(\alpha, \beta, \gamma)$ is the foot of the perpendicular drawn from the point $M(1, 0, 1)$ to P , then $3(\alpha + \beta + \gamma)$ equals **03 Sep 2020 (E)**

Q226*. The plane passing through the points $(1, 2, 1)$, $(2, 1, 2)$ and parallel to the line, $2x = 3y$, $z = 1$ also passes through the point **02 Sep 2020 (M)**

(1) $(0, 6, -2)$ (2) $(-2, 0, 1)$
(3) $(0, -6, 2)$ (4) $(2, 0, -1)$

Q227*. A plane passing through the point $(3, 1, 1)$ contains two lines whose direction ratios are $1, -2, 2$ and $2, 3, -1$ respectively. If, this plane also passes through the point $(\alpha, -3, 5)$, then α is equal to **02 Sep 2020 (E)**

(1) 5 (2) -10
(3) 10 (4) -5

Q228. The projection of the line segment joining the point $(1, -1, 3)$ and $(2, -4, 11)$ on the line joining the points $(-1, 2, 3)$ and $(3, -2, 10)$ is **09 Jan 2020 (M)**

Q229*. If the distance between the plane, $23x - 10y - 2z + 48 = 0$ and the plane containing the lines $\frac{x+1}{2} = \frac{y-3}{4} = \frac{z+1}{3}$ and $\frac{x+3}{2} = \frac{y+2}{6} = \frac{z-1}{\lambda}$ ($\lambda \in R$) is equal to $\frac{k}{\sqrt{633}}$, then k is equal to **09 Jan 2020 (E)**

Q230. The shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is **08 Jan 2020 (M)**

- (1) $2\sqrt{30}$ (2) $\frac{7}{2}\sqrt{30}$
(3) $3\sqrt{30}$ (4) 3

Q231*. The mirror image of the point $(1, 2, 3)$, in a plane is $(-\frac{7}{3}, -\frac{4}{3}, -\frac{1}{3})$. Which of the following points lies on this plane? **08 Jan 2020 (E)**

- (1) $(1, 1, 1)$ (2) $(1, -1, 1)$
(3) $(-1, -1, 1)$ (4) $(-1, -1, -1)$

Q232. If the foot of the perpendicular drawn from the point $(1, 0, 3)$ on a line passing through $(\alpha, 7, 1)$ is $(\frac{5}{3}, \frac{7}{3}, \frac{17}{3})$, then α is equal to **07 Jan 2020 (E)**

Q233*. Let P be a plane passing through the points $(2, 1, 0)$, $(4, 1, 1)$ and $(5, 0, 1)$ and R be any point $(2, 1, 6)$. Then the image of R in the plane P is **07 Jan 2020 (M)**

- (1) $(6, 5, 2)$ (2) $(6, 5, -2)$
(3) $(4, 3, 2)$ (4) $(3, 4, -2)$

Q234*. A plane which bisects the angle between the two given planes $2x - y + 2z - 4 = 0$ and $x + 2y + 2z - 2 = 0$, passes through the point **12 Apr 2019 (E)**

- (1) $(2, 4, 1)$ (2) $(1, -4, 1)$
(3) $(1, 4, -1)$ (4) $(2, -4, 1)$

Q235*. The length of the perpendicular drawn from the point $(2, 1, 4)$ to the plane containing the lines $\vec{r} = (\hat{i} + \hat{j}) + \lambda(\hat{i} + 2\hat{j} - \hat{k})$ and $\vec{r} = (\hat{i} + \hat{j}) + \mu(-\hat{i} + \hat{j} - 2\hat{k})$ is **12 Apr 2019 (E)**

- (1) $\frac{1}{3}$ (2) 3
(3) $\sqrt{3}$ (4) $\frac{1}{\sqrt{3}}$

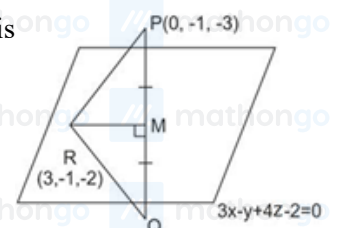
Q236*. If the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the plane $2x + 3y - z + 13 = 0$ at a point P and the plane $3x + y + 4z = 16$ at a point Q , then PQ is equal to **12 Apr 2019 (M)**

- (1) $2\sqrt{7}$ (2) 14
(3) $2\sqrt{14}$ (4) $\sqrt{14}$

Q237. If the length of the perpendicular from the point $(\beta, 0, \beta)$, $(\beta \neq 0)$ to the line, $\frac{x}{1} = \frac{y-1}{0} = \frac{z+1}{-1}$ is $\sqrt{\frac{3}{2}}$, then β is equal to **10 Apr 2019 (M)**

- (1) 2 (2) -1
(3) -2 (4) 1

Q238*. If $Q(0, -1, -3)$ is the image of the point P in the plane $3x - y + 4z = 2$ and R is the point $(3, -1, -2)$, then the area (in sq. units) of ΔPQR is



10 Apr 2019 (M)

(1) $\frac{\sqrt{91}}{4}$
(3) $2\sqrt{13}$

(2) $\frac{\sqrt{91}}{2}$
(4) $\frac{\sqrt{65}}{2}$

Q239*. If the plane $2x - y + 2z + 3 = 0$ has the distances $\frac{1}{3}$ and $\frac{2}{3}$ units from the planes $4x - 2y + 4z + \lambda = 0$ and $2x - y + 2z + \mu = 0$, respectively, then the maximum value of $\lambda + \mu$ is equal to: **10 Apr 2019 (E)**

- (1) 9
(2) 15
(3) 13
(4) 5

Q240*. A perpendicular is drawn from a point on the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1}$ to the plane $x + y + z = 3$ such that the foot of the perpendicular Q also lies on the plane $x - y + z = 3$. Then the coordinates of Q are **10 Apr 2019 (E)**

- (1) (2, 0, 1)
(2) (-1, 0, 4)
(3) (4, 0, -1)
(4) (1, 0, 2)

Q241*. If the line, $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$ meets the plane, $x + 2y + 3z = 15$ at a point P , then the distance of P from the origin is, **09 Apr 2019 (M)**

- (1) $2\sqrt{5}$
(2) $\frac{9}{2}$
(3) $\frac{\sqrt{5}}{2}$
(4) $\frac{7}{2}$

Q242. The vertices B and C of a ΔABC lie on the line, $\frac{x+2}{3} = \frac{y-1}{0} = \frac{z}{4}$ such that $BC = 5$ units. Then the area (in sq. units) of this triangle, given the point $A(1, -1, 2)$, is **09 Apr 2019 (E)**

- (1) 6
(2) $2\sqrt{34}$
(3) $\sqrt{34}$
(4) $5\sqrt{17}$

Q243*. A plane passing through the points $(0, -1, 0)$ and $(0, 0, 1)$ and making an angle $\frac{\pi}{4}$ with the plane $y - z + 5 = 0$, also passes through the point **09 Apr 2019 (M)**

- (1) $(\sqrt{2}, -1, 4)$
(2) $(\sqrt{2}, 1, 4)$
(3) $(-\sqrt{2}, -1, -4)$
(4) $(-\sqrt{2}, 1, -4)$

Q244*. Let P be the plane, which contains the line of intersection of the planes, $x + y + z - 6 = 0$ and $2x + 3y + z + 5 = 0$ and it is perpendicular to the xy -plane. Then the distance of the point $(0, 0, 256)$ from P is equal to: **09 Apr 2019 (E)**

- (1) $205\sqrt{5}$ units
(2) $\frac{17}{\sqrt{5}}$ units
(3) $\frac{11}{\sqrt{5}}$ units
(4) $63\sqrt{5}$ units

Q245. The length of the perpendicular from the point $(2, -1, 4)$ on the straight line $\frac{x+3}{10} = \frac{y-2}{-7} = \frac{z}{1}$ is **08 Apr 2019 (M)**

- (1) greater than 3 but less than 4
(2) greater than 4
(3) less than 2
(4) greater than 2 but less than 3

Q246*. The vector equation of the plane through the line of intersection of the planes $x + y + z = 1$ and $2x + 3y + 4z = 5$ which is perpendicular to the plane $x - y + z = 0$ is **08 Apr 2019 (E)**

$$(1) \vec{r} \times (\hat{i} + \hat{k}) + 2 = 0$$

$$(3) \vec{r} \times (\hat{i} - \hat{k}) + 2 = 0$$

$$(2) \vec{r} \cdot (\hat{i} - \hat{k}) - 2 = 0$$

$$(4) \vec{r} \cdot (\hat{i} - \hat{k}) + 2 = 0$$

Q247. If a point $R(4, y, z)$ lies on the line segment joining the points $P(2, -3, 4)$ and $Q(8, 0, 10)$, then the distance of R from the origin is **08 Apr 2019 (E)**

$$(1) 2\sqrt{21}$$

$$(2) \sqrt{53}$$

$$(3) 6$$

$$(4) 2\sqrt{14}$$

Q248*. The equation of a plane containing the line of intersection of the planes $2x - y - 4 = 0$ and $y + 2z - 4 = 0$ and passing through the point $(1, 1, 0)$ is **08 Apr 2019 (M)**

$$(1) x - 3y - 2z = -2$$

$$(2) x + 3y + z = 4$$

$$(3) x - y - z = 0$$

$$(4) 2x - z = 2$$

Q249*. If an angle between the line, $\frac{x+1}{2} = \frac{y-2}{1} = \frac{z-3}{-2}$ and the plane, $x - 2y - kz = 3$ is $\cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$, then a value of k is **12 Jan 2019 (E)**

$$(1) \sqrt{\frac{5}{3}}$$

$$(2) \sqrt{\frac{3}{5}}$$

$$(3) -\frac{3}{5}$$

$$(4) -\frac{5}{3}$$

Q250*. A tetrahedron has vertices $P(1, 2, 1)$, $Q(2, 1, 3)$, $R(-1, 1, 2)$ and $O(0, 0, 0)$. The angle between the faces OPQ and PQR is **12 Jan 2019 (M)**

$$(1) \cos^{-1}\left(\frac{7}{31}\right)$$

$$(2) \cos^{-1}\left(\frac{17}{31}\right)$$

$$(3) \cos^{-1}\left(\frac{19}{35}\right)$$

$$(4) \cos^{-1}\left(\frac{9}{35}\right)$$

Q251*. The perpendicular distance from the origin to the plane containing the two lines, $\frac{x+2}{3} = \frac{y-2}{5} = \frac{z+5}{7}$ and $\frac{x-1}{1} = \frac{y-4}{4} = \frac{z+4}{7}$, is **12 Jan 2019 (M)**

$$(1) 11\sqrt{6}$$

$$(2) \frac{11}{\sqrt{6}}$$

$$(3) 11$$

$$(4) 6\sqrt{11}$$

Q252*. Let S be the set of all real values of λ such that a plane passing through the points $(-\lambda^2, 1, 1)$, $(1, -\lambda^2, 1)$ and $(1, 1, -\lambda^2)$ also passes through the point $(-1, -1, 1)$. Then S is equal to : **12 Jan 2019 (E)**

$$(1) \{\sqrt{3}\}$$

$$(2) \{3, -3\}$$

$$(3) \{1, -1\}$$

$$(4) \{\sqrt{3}, -\sqrt{3}\}$$

Q253. The direction ratios of normal to the plane through the points $(0, -1, 0)$ and $(0, 0, 1)$ and making an angle $\frac{\pi}{4}$ with the plane $y - z + 5 = 0$ are; $2, -1, 1$ $2, \sqrt{2} - \sqrt{2}, 1, -1$ $2\sqrt{3}, 1, -1$ **11 Jan 2019 (M)**

$$(1) \text{option 1 and 2}$$

$$(2) \text{option 2 and 3}$$

$$(3) \text{option 3 and 4}$$

$$(4) \text{all the options}$$

Q254. Two lines $\frac{x-3}{1} = \frac{y+1}{3} = \frac{z-6}{-1}$ and $\frac{x+5}{7} = \frac{y-2}{-6} = \frac{z-3}{4}$ intersect at the point R . The reflection of R in the xy -plane has coordinates: **11 Jan 2019 (E)**

$$(1) (2, -4, -7)$$

$$(2) (2, 4, 7)$$

$$(3) (2, -4, 7)$$

$$(4) (-2, 4, 7)$$

Q255*. If the point $(2, \alpha, \beta)$ lies on the plane which passes through the points $(3, 4, 2)$ and $(7, 0, 6)$ and is perpendicular to the plane $2x - 5y = 15$, then $2\alpha - 3\beta$ is equal to : 11 Jan 2019 (E)

- (1) 12 (2) 7
(3) 5 (4) 17

Q256*. The plane containing the line $\frac{x-3}{2} = \frac{y+2}{-1} = \frac{z-1}{3}$ and also containing its projection on the plane $2x + 3y - z = 5$, contains which one of the following points? 11 Jan 2019 (M)

- (1) $(2, 2, 0)$ (2) $(-2, 2, 2)$
(3) $(0, -2, 2)$ (4) $(2, 0, -2)$

Q257*. The plane passing through the point $(4, -1, 2)$ and parallel to the lines $\frac{x+2}{3} = \frac{y-2}{-1} = \frac{z+1}{2}$ and $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{3}$ also passes through the point 10 Jan 2019 (M)

- (1) $(1, 1, -1)$ (2) $(-1, -1, -1)$
(3) $(-1, -1, 1)$ (4) $(1, 1, 1)$

Q258*. The plane which bisects the line segment joining the points $(-3, -3, 4)$ and $(3, 7, 6)$ at right angles, passes through which one of the following points? 10 Jan 2019 (E)

- (1) $(2, 1, 3)$ (2) $(4, 1, -2)$
(3) $(4, -1, 7)$ (4) $(-2, 3, 5)$

Q259*. On which of the following lines lies the point of intersection of the line, $\frac{x-4}{2} = \frac{y-5}{2} = \frac{z-3}{1}$ and the plane, $x + y + z = 2$? 10 Jan 2019 (E)

- (1) $\frac{x-4}{1} = \frac{y-5}{1} = \frac{z-3}{-1}$ (2) $\frac{x-1}{1} = \frac{y-3}{2} = \frac{z+4}{-5}$
(3) $\frac{x-2}{2} = \frac{y-3}{2} = \frac{z+3}{3}$ (4) $\frac{x+3}{3} = \frac{4-y}{3} = \frac{z+1}{-2}$

Q260*. The equation of the line passing through $(-4, 3, 1)$, parallel to the plane $x + 2y - z - 5 = 0$ and intersecting the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z-2}{-1}$ is 09 Jan 2019 (M)

- (1) $\frac{x+4}{3} = \frac{y-3}{-1} = \frac{z-1}{1}$ (2) $\frac{x+4}{1} = \frac{y-3}{1} = \frac{z-1}{3}$
(3) $\frac{x+4}{-1} = \frac{y-3}{1} = \frac{z-1}{1}$ (4) $\frac{x-4}{2} = \frac{y+3}{1} = \frac{z+1}{4}$

Q261*. The plane through the intersection of the planes $x + y + z = 1$ and $2x + 3y - z + 4 = 0$ and parallel to y -axis also passes through the point 09 Jan 2019 (M)

- (1) $(3, 3, -1)$ (2) $(-3, 1, 1)$
(3) $(3, 2, 1)$ (4) $(-3, 0, -1)$

Q262*. The equation of the plane containing the straight line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ and $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$ is: 09 Jan 2019 (E)

- (1) $3x + 2y - 3z = 0$ (2) $x + 2y - 2z = 0$
(3) $x - 2y + z = 0$ (4) $5x + 2y - 4z = 0$

Q263*. If the lines $x = ay + b, z = cy + d$ and $x = a'z + b', y = c'z + d'$ are perpendicular, then 09 Jan 2019 (E)

- (1) $cc' + a + a' = 0$ (2) $aa' + c + c' = 0$
(3) $bb' + cc' + 1 = 0$ (4) $ab' + bc' + 1 = 0$

ANSWER KEYS

1. (2)	2. (62)	3. (1)	4. (4)	5. (4)	6. (4)	7. (11)	8. (1)
9. (2)	10. (13)	11. (2)	12. (43)	13. (4)	14. (4)	15. (3)	16. (25)
17. (6)	18. (3)	19. (2)	20. (48)	21. (2)	22. (3)	23. (21)	24. (3)
25. (2)	26. (4)	27. (3)	28. (22)	29. (12)	30. (2)	31. (16)	32. (1)
33. (196)	34. (4)	35. (2)	36. (3)	37. (9)	38. (65)	39. (4)	40. (2)
41. (2)	42. (2)	43. (108)	44. (3)	45. (1)	46. (1)	47. (26)	48. (5)
49. (4)	50. (15)	51. (1)	52. (2)	53. (3)	54. (3)	55. (3)	56. (4)
57. (2)	58. (1)	59. (11)	60. (5)	61. (4)	62. (4)	63. (4)	64. (3)
65. (3)	66. (1)	67. (7)	68. (2)	69. (4)	70. (9)	71. (1)	72. (2)
73. (1)	74. (11)	75. (18)	76. (2)	77. (4)	78. (1)	79. (594)	80. (1)
81. (3)	82. (6)	83. (4)	84. (1)	85. (10)	86. (4)	87. (3)	88. (1)
89. (2)	90. (180)	91. (9)	92. (15)	93. (315)	94. (3)	95. (4)	96. (158)
97. (1)	98. (4)	99. (2)	100. (2)	101. (9)	102. (355)	103. (1)	104. (4)
105. (1)	106. (1)	107. (3)	108. (18)	109. (9)	110. (14)	111. (3)	112. (3)
113. (384)	114. (4)	115. (4)	116. (2)	117. (2)	118. (10)	119. (450)	120. (4)
121. (2)	122. (2)	123. (2)	124. (4)	125. (2)	126. (12)	127. (1)	128. (1)
129. (2)	130. (125)	131. (153)	132. (1)	133. (3)	134. (3)	135. (3)	136. (2)
137. (26)	138. (28)	139. (2)	140. (2)	141. (125)	142. (2)	143. (1)	144. (4)
145. (2)	146. (3)	147. (1)	148. (1)	149. (137)	150. (2)	151. (3)	152. (2)
153. (1)	154. (51)	155. (5)	156. (2)	157. (3)	158. (1)	159. (84)	160. (2)
161. (3)	162. (2)	163. (2)	164. (3)	165. (4)	166. (61)	167. (7)	168. (4)
169. (2)	170. (4)	171. (4)	172. (72)	173. (1)	174. (2)	175. (96)	176. (26)
177. (4)	178. (7)	179. (4)	180. (3)	181. (1)	182. (3)	183. (1)	184. (2)
185. (6)	186. (1)	187. (4)	188. (81)	189. (28)	190. (4)	191. (1)	192. (38)
193. (4)	194. (4)	195. (0)	196. (2)	197. (2)	198. (4)	199. (2)	200. (2)
201. (2)	202. (3)	203. (4)	204. (8)	205. (2)	206. (1)	207. (3)	208. (44)
209. (1)	210. (4)	211. (4)	212. (1)	213. (3)	214. (1)	215. (1)	216. (2)
217. (3)	218. (1)	219. (2)	220. (3)	221. (2)	222. (1)	223. (1)	224. (2)
225. (5)	226. (2)	227. (1)	228. (8)	229. (3)	230. (3)	231. (2)	232. (4)
233. (2)	234. (4)	235. (3)	236. (3)	237. (2)	238. (2)	239. (3)	240. (1)
241. (2)	242. (3)	243. (2)	244. (3)	245. (1)	246. (4)	247. (4)	248. (3)
249. (1)	250. (3)	251. (2)	252. (4)	253. (2)	254. (1)	255. (2)	256. (4)
257. (4)	258. (2)	259. (2)	260. (1)	261. (3)	262. (3)	263. (2)	

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